

The search budget of the BSM resonance program

How many independent bump hunts the model space implies,
and what a discovery in that program costs

T. Reisch
Université de Genève

August 11, 2026

Abstract. Searches for new resonances are increasingly run as broad scans over many mass spectra rather than as single model-specific measurements. For a scan of that kind the first number one needs is not a limit but a count: how many independent looks does the program contain, and what local significance does a discovery in it therefore have to reach? This note works that number out from published information alone. Organising the public BSM resonance space by the spectrum in which each model peaks, rather than by the theory that predicts it, collapses several dozen model classes onto **46 distinct invariant-or transverse-mass spectra**, with lepton flavour and b -tag content counted as part of the mass axis. Counting Gross–Vitells resolution elements over the published scan window of each spectrum gives $N_{\text{trials}} \approx 3.7 \times 10^3$ inclusively and 6.6×10^3 at realistic event-selection granularity, so a 5σ global discovery costs a local significance of $Z_{\text{local}} \approx 6.4$ to 6.5 . That figure is stable to $\pm 0.1\sigma$ against every counting choice we varied, since the penalty enters only as $\ln N_{\text{trials}}$, and the same arithmetic reproduces the observed population of ATLAS excesses (tens at 3σ , none spurious at 5σ). We then price a fully combinatorial scan ($N_{\text{trials}} \approx 1.4 \times 10^5$, $Z_{\text{local}} \approx 7.0$), show that 57 of the 82 reachable object compositions have never been scanned by any published ATLAS search, compare three candidate-selection rules on a common ROC, and evaluate a two-stage A/B unblinding scheme that costs about 0.5σ in reach and buys an exactly countable trials factor.

Contents

1	Introduction and scope	3
1.1	The question	3
1.2	The one-paragraph answer	3
1.3	What this note does and does not use	3
1.4	Conventions	4
2	The search budget	4
2.1	Method	4
2.2	Result	5
2.3	Which models feed which spectrum	5
2.4	How finely should searches be counted?	5
2.5	Reading the result	5
3	The combinatorial scan, and what has never been looked at	9
3.1	Budget of a fully combinatorial scan	9
3.2	Which compositions has ATLAS actually scanned?	9
3.3	Three reasons the covered fraction is overstated	10
3.4	The programme that was meant to cover the rest	10
3.5	Named gaps	11
4	Does the arithmetic reproduce the excesses we have seen?	11
4.1	The prediction	12
4.2	The record	12
4.3	Verdict	13
5	Selecting candidates from a scan	13
5.1	The background maximum is not Gaussian	13
5.2	A real signal often loses the scan	14
5.3	A threshold dominates the argmax	15
5.4	Letting the data set the cut: Benjamini–Hochberg	16
5.5	The same three rules with an imperfect estimator	16
5.6	What this implies for a data-directed scan	18
6	Two-stage A/B unblinding	18
6.1	The proposal	18
6.2	The cost	19
6.3	The look-elsewhere effect is conserved, not reduced	20
6.4	Would swapping the halves recover it?	20
6.5	When the split is nonetheless the right choice	21
6.6	The failure mode splitting does <i>not</i> address	22
6.7	What changes if the significance estimator is imperfect	22
6.8	Recommended working point	26
7	Software and reproducibility	26
7.1	The two modules everything rests on	26
7.2	Layout and rebuilding	26
7.3	Determinism	27
7.4	Relation to the catalogue note	27

8	Summary and recommendations	27
8.1	What was established	27
8.2	Recommendations	28
8.3	Known limitations	28
A	Scan windows and their sources	29

1 Introduction and scope

1.1 The question

A conventional search fixes a model, a final state and a mass window before looking at data. The look-elsewhere effect is then a modest correction applied at the end, and nobody worries much about it. A growing fraction of the resonance program does not work that way. Model-independent and data-directed scans [1], machine-learning anomaly searches [2] and general searches over many event classes [3] all sweep large parts of the mass space at once, and for those the trials factor is not a footnote. It is the leading term in what a discovery has to clear.

A prior question therefore needs a careful answer: how many independent looks does the BSM resonance program actually contain? The number is usually quoted per analysis, if at all, and never for the field as a whole. This note works it out from the ground up.

1. How many independent bump hunts does covering the public BSM model space imply, and what local significance does a 5σ global discovery therefore cost (Section 2)?
2. What does the same arithmetic say about a fully combinatorial scan, and which object combinations has ATLAS in fact never scanned (Section 3)?
3. Is the resulting trials factor consistent with the excesses ATLAS has actually reported (Section 4)?
4. Given a scan, how should candidates be selected: the largest bin, a threshold, or a false-discovery-rate rule (Section 5)?
5. Does a two-stage A/B unblinding scheme make the trials factor cheaper, or only more defensible (Section 6)?

1.2 The one-paragraph answer

The BSM resonance space looks unbounded at the level of models, with dozens of theories and thousands of signal samples across them, and it collapses to **46 distinct invariant- or transverse-mass spectra** once organised by where the bump appears, with lepton flavour and b -tag content counted as part of the mass axis. Counting Gross–Vitells resolution elements [4] over each spectrum’s published scan window, the whole program amounts to $N_{\text{trials}} \approx 3.7 \times 10^3$ independent looks, rising to 6.6×10^3 when searches are sliced at realistic event-selection granularity. Because the penalty enters as $\ln N_{\text{trials}}$, that program costs a local bar of $Z_{\text{local}} \approx 6.4$ to 6.5 for a 5σ global discovery, and the figure barely moves under any counting choice we varied. **Breadth is cheap.** What is expensive is anything that makes N_{trials} uncountable, which is the situation a neural-network-driven scan naturally creates, and the reason the two-stage unblinding discussion in Section 6 matters even though splitting the data cannot reduce the look-elsewhere effect.

1.3 What this note does and does not use

Every number here derives from two data-free ingredients: a curated list of bump observables with their published scan windows, floors and resolutions (Appendix A), and a map from public BSM model classes to the spectra they populate. Both live in one module each in the accompanying repository (Section 7), and neither reads a single experiment-internal input — no dataset identifiers, no signal-sample inventory, no unpublished analysis configuration. Whether a given spectrum happens to have centrally produced Monte-Carlo samples behind it is a question about a collaboration’s production program rather than about the search budget, and it is deliberately outside the scope of this note; a companion ATLAS-internal note treats it.

One consequence is worth stating early, because it is what keeps the exercise non-circular. Every spectrum is counted over the mass range a *published* search family scans, never over a range inferred from which signal samples exist. A search does not become cheaper because few samples were generated for it.

1.4 Conventions

Throughout, one *search* means one scanned mass spectrum, n_s is the number of effective independent resolution elements it contains, and $N_{\text{trials}} = \sum_s n_s$. The local significance required for a 5σ global discovery follows from

$$Z_{\text{global}}^2 = Z_{\text{local}}^2 - 2 \ln N_{\text{trials}} \quad \implies \quad Z_{\text{local}} = \sqrt{25 + 2 \ln N_{\text{trials}}}, \quad (1)$$

the asymptotic Gross–Vitells relation [4] in the fixed-resolution-element approximation. Masses are in GeV unless stated otherwise, and $\sqrt{s} = 13.6$ TeV.

Bibliographic details for the external searches quoted throughout were checked against the arXiv metadata records and INSPIRE in August 2026.

2 The search budget

2.1 Method

One bump hunt is one scanned mass spectrum. For a spectrum with constant fractional mass resolution $\sigma_M = r M$, the number of effectively independent resolution elements across a scanned window $[M_{\text{lo}}, M_{\text{hi}}]$ is

$$n_s = \frac{1}{r} \ln \frac{M_{\text{hi}}}{M_{\text{lo}}}, \quad (2)$$

summed over disjoint segments where a scan is not contiguous. The program’s trials factor is $N_{\text{trials}} = \sum_s n_s$ and the discovery bar follows from Eq. (1). This is the fixed-resolution-element form of the Gross–Vitells estimate [4]; the up-crossing refinement adds a mild Z -dependence that is immaterial at the precision quoted here.

The fractional resolution r is the only genuine physics input, so every headline is quoted with a band obtained by varying r by a factor two in each direction, which moves N_{trials} by a factor two the other way.

The window convention, and why it matters. Which range counts as “scanned” is the one choice that can quietly bias the whole exercise. We count every spectrum over its **published-search scan window**, meaning the mass range a real, published ATLAS search family scans on that axis, extended where that family’s published signal grid demonstrably prepares a wider scan. Each window carries a per-channel source note (Appendix A).

The tempting alternative — taking the range over which signal samples happen to have been generated — is wrong twice over. It understates sparsely sampled channels, since a search does not become cheaper because few samples were made for it, and it ties a supposedly model-space quantity to one collaboration’s production choices. With the published-window convention the budget is a statement about the searches the field has defined.

Scope. Only invariant- and transverse-mass bump hunts enter the sum. The dedicated non-bump programs, meaning displaced HNL, Z_d , RPV and ALP searches, MET+jet, dE/dx monopole and off-shell EFT, carry their own look-elsewhere effect and are not mass scans. They are excluded rather than summed.

What counts as one spectrum. Models are grouped by the mass axis in which they peak, and two properties that a real analysis must select on are treated as part of that axis rather than as cuts within it: **lepton flavour** and **b -tag content**. So $m(ee)$ and $m(\mu\mu)$, and likewise $m(eb)$ and $m(ej)$, are separate spectra, each with its own resolution and scan window. The justification is physical rather than bookkeeping: an electron pair is measured in the electromagnetic calorimeter ($r \approx 0.015$, nearly mass-independent) while a muon pair is measured by track sagitta (r rising to $\sim 15\%$ at 3 TeV, entered as a window-averaged $r \approx 0.05$), and the two are scanned by separate analyses with separate triggers. At low mass the ordering reverses, which is why the dark-photon electron channel starts at 1 GeV and the muon channel at 0.3 GeV. Because flavour is an axis here, it is *not* also counted in the event-selection multiplicities below; doing both would double-count it.

Conversely, decay modes that resolve onto the *same* mass axis are merged: $X \rightarrow HH$ observed in $4b$ is not a different spectrum from $X \rightarrow HH$, it is a different event selection of it, and it is counted as such in Table 3.

2.2 Result

Table 1 gives the per-spectrum breakdown and Table 2 the totals. Covering every bump channel that public BSM models motivate costs

$$N_{\text{trials}} = 3694 \quad \text{over 46 spectra} \quad \implies \quad Z_{\text{local}} = 6.44 \text{ for a } 5\sigma \text{ global discovery.}$$

Equivalently, a local 5σ anywhere in that program is worth about 3σ globally.

2.3 Which models feed which spectrum

The mapping from model classes to spectra is the other half of the input, and it is worth being able to see it rather than trust it. Figure 2 draws it in full. Two structural features matter for the budget. First, the matrix is far from diagonal: popular spectra such as $m(jj)$, $m(VV)$ and $m(t\bar{t})$ are motivated by eight to twelve model classes each, which is exactly why the number of *searches* is so much smaller than the number of *models*. Second, adding a model to a spectrum that is already scanned costs *nothing* in trials. The entire cost of breadth sits in opening new columns, never in filling in existing ones.

2.4 How finely should searches be counted?

An inclusive count of one spectrum per observable is the lower bound. Real searches slice each final state further, and each slice scanned for a bump is another look. Published searches split by b -tag category, boosted versus resolved regime, sub-decay mode, 0/1/2-lepton bin and ambiguous pairing — a leptoquark pair search scans all four lepton-jet pairings. Attaching a documented, publication-anchored selection multiplicity to each observable gives 94 channels and $N_{\text{trials}} = 6616$, hence $Z_{\text{local}} = 6.53$. Lepton flavour is deliberately *not* one of the multipliers, since it is already a mass axis.

2.5 Reading the result

The headline is robust by construction. N_{trials} enters only through $\ln N_{\text{trials}}$, so a factor two error in the resolution assumption, or a factor 1.8 in how finely channels are counted, moves the discovery bar by at most $\pm 0.1\sigma$. Every granularity level of this study in Table 3 sits within 0.05σ of 6.5.

Two conclusions follow, and they pull in opposite directions.

Breadth is cheap. Covering the entire known BSM resonance space costs about one extra unit of σ relative to a single pre-specified counting experiment, 6.5 against 5. There is no

Table 1: Trials per spectrum. r is the assumed fractional mass resolution, the window is the published-search scan range (Appendix A) and n_s follows from Eq. (2). “envelope” is the reference count over the full kinematically accessible range (analyzable floor to $\sqrt{s}$), “#mod” the number of public model classes that predict a peak in the spectrum, and “#sel” the number of distinct published event selections that scan it (Table 3).

#	spectrum	r	window [GeV]	n_s	envelope	#mod	#sel	$n_s \times \text{sel}$
1	$m(\gamma\gamma)$	0.01	60–110 + 150–5000	411	583	5	2	822
2	$m(\mu\mu)$ (Z_d)	0.02	0.3–400	360	536	2	1	360
3	$m(ee)$	0.015	150–8000	265	321	7	1	265
4	$m(ee)$ (Z_d)	0.025	1–400	240	381	1	1	240
5	$m(e\gamma)$	0.015	200–5000	215	328	1	1	215
6	$m(e\mu)$ LFV	0.03	100–8000	146	234	4	1	146
7	$m(e^\pm e^\pm)$ SS	0.015	200–1300	125	454	2	1	125
8	$m(V\gamma)$	0.03	220–6000	110	141	3	2	220
9	$m(ej)$	0.03	200–5000	107	234	3	4	429
10	$m(\mu\gamma)$	0.03	200–5000	107	164	1	1	107
11	$m(\mu j)$	0.04	200–5000	80	176	3	4	322
12	$m(eZ)$	0.03	100–1100	80	164	1	1	80
13	$m(\mu\mu)$	0.05	150–8000	80	96	7	1	80
14	$m(cb)$ dijet	0.05	20–1000	78	136	1	2	156
15	$m(e^\pm \mu^\pm)$ SS	0.025	200–1300	75	272	2	1	75
16	$m(jj)$	0.05	200–8000	74	84	8	2	148
17	$m(j\gamma)$	0.04	500–7000	66	113	2	1	66
18	$m(\mu Z)$	0.04	100–1100	60	123	1	1	60
19	$m(eb)$	0.04	200–2000	58	170	3	2	115
20	$m(eejj)$	0.05	400–7000	57	76	2	1	57
21	$m(VV)$	0.06	200–6000	57	74	12	9	510
22	$m(bj)$	0.06	200–6000	57	70	1	2	113
23	$m(Vh)$	0.06	200–5000	54	70	2	6	322
24	$m(\mu^\pm \mu^\pm)$ SS	0.04	200–1300	47	170	2	1	47
25	$m(\mu b)$	0.05	200–2000	46	136	3	2	92
26	multilepton	0.12	50–10000	44	62	3	4	177
27	$m(3j)$	0.05	200–1800	44	84	2	1	44
28	$m(tb)$	0.08	180–6000	44	54	3	3	132
29	$m(\tau\tau)$	0.12	35–6000	43	50	4	3	129
30	$m(\mu\mu jj)$	0.07	400–7000	41	54	2	1	41
31	$m(bb)$	0.1	15–62 + 450–6000	40	68	4	3	120
32	$m(HH)$	0.08	250–6000	40	50	4	6	238
33	$m_T(e\nu)$	0.1	150–7000	38	60	2	1	38
34	$m(t\bar{t})$	0.08	350–6000	36	46	9	4	142
35	$m(\tau j)$	0.1	200–5000	32	56	3	2	64
36	$m(e\tau)$ LFV	0.12	100–3000	28	50	4	1	28
37	$m(\mu\tau)$ LFV	0.12	100–3000	28	50	4	1	28
38	$m_T(\tau\nu)$	0.12	200–5000	27	47	1	1	27
39	$m_T(\mu\nu)$	0.15	150–7000	26	40	2	1	26
40	$m(tW)$	0.08	500–3000	22	50	1	2	45
41	$m(\text{multi})$	0.06	3000–10500	21	64	2	1	21
42	$m(t\bar{t})/m(jj)$ (sgluon)	0.08	400–2000	20	53	2	2	40
43	$m(\tau b)$	0.12	200–2000	19	47	3	2	38
44	$m(t\bar{t}Z)/m(Zt)$	0.08	1000–4000	17	43	1	2	35
45	$m(Wb)$	0.08	800–3000	16	61	1	1	16
46	$m(Ht)$	0.08	1000–3000	14	48	1	1	14
total (46 spectra)				3 694	6 442		94	6 616

Table 2: Search-budget summary. The band comes from varying the fractional resolution r by a factor two in each direction.

estimate	spectra	N_{trials}	band	Z_{local}	band
inclusive (one spectrum per observable)	46	3694	1847–7388	6.44	6.33–6.54
published event selections	94	6616	3308–13232	6.53	6.42–6.63
kinematic envelope (reference bound)	46	6442	3221–12884	6.52	6.42–6.63

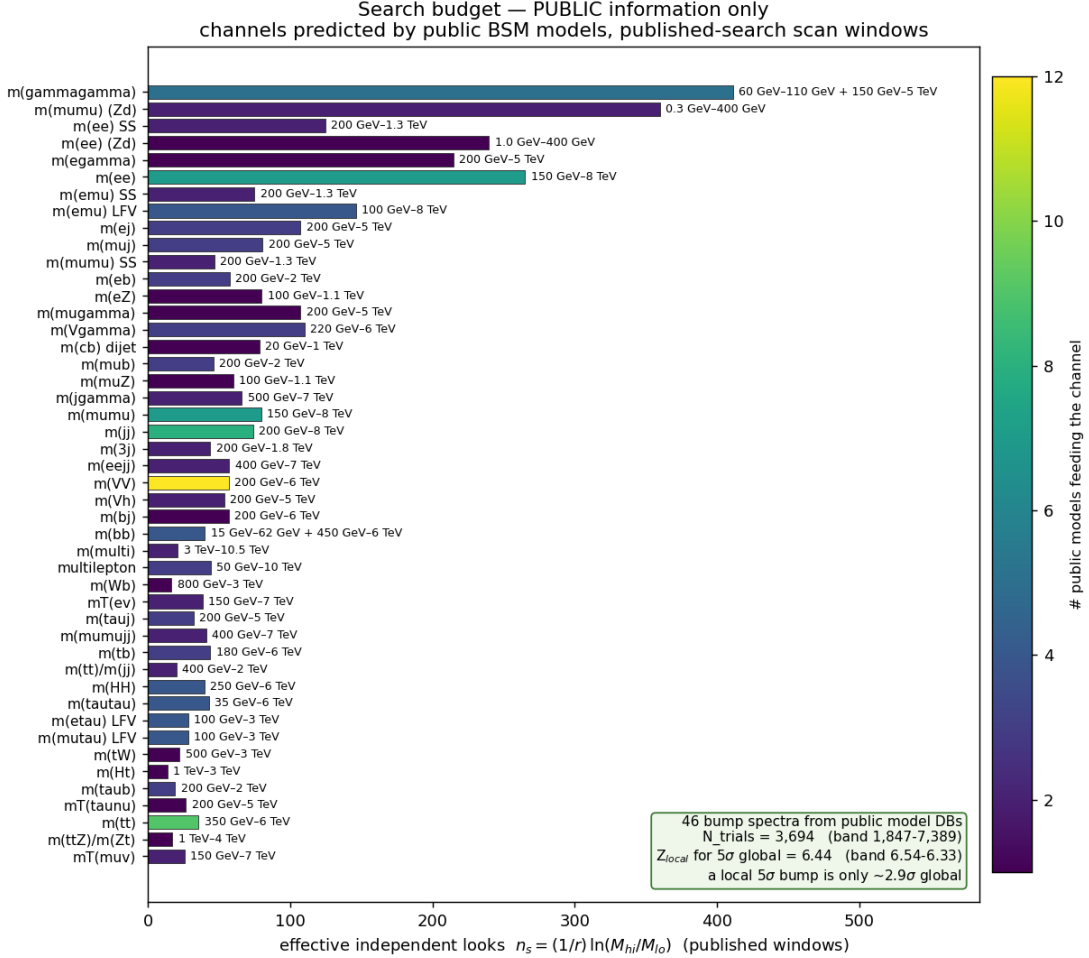


Figure 1: Trials per spectrum, with the scan window annotated and the colour giving the number of public model classes that feed the channel. Note the strong hierarchy: the five highest-resolution spectra ($m(\gamma\gamma)$, the two dark-photon axes, $m(ee)$ and $m(e\gamma)$) carry 40 % of the entire budget, simply because $n_s \propto 1/r$.

Table 3: The budget at increasing granularity.

granularity	spectra	N_{trials}	Z_{local}
inclusive (one spectrum per observable)	46	3694	6.44
real event selections	94	6616	6.53
full ATLAS BSM program (literature, Section 4)	—	$\sim 5 \times 10^4$	6.83

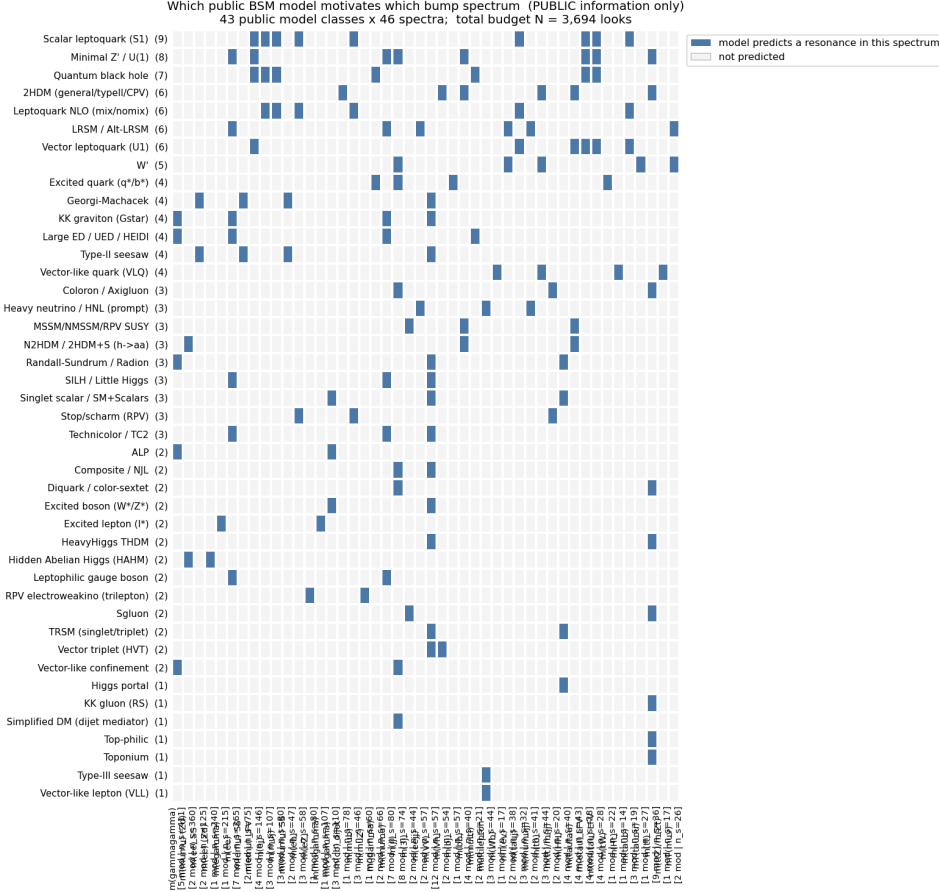


Figure 2: Public model classes (rows) against canonical bump spectra (columns). A filled cell means the model predicts a peak in that spectrum. The matrix is derived from the model map rather than asserted by hand.

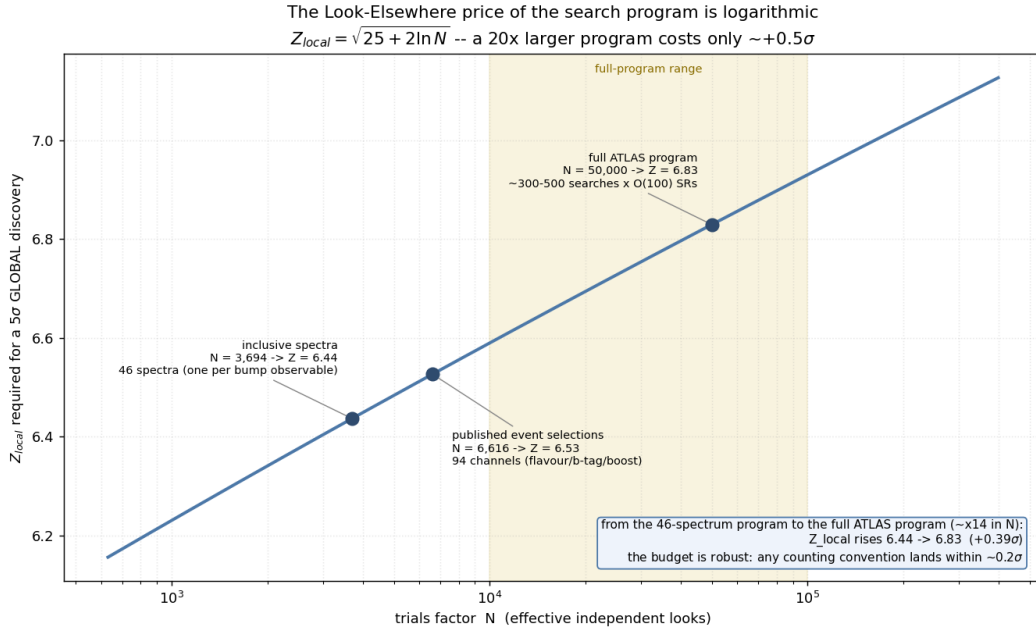


Figure 3: Z_{local} required for a 5σ global discovery as a function of the trials factor, with the granularity levels of this study placed on the curve. A factor 1.8 in channel counting moves the discovery bar by 0.09σ , and reaching the full ATLAS program, a further factor ~ 8 , adds only another 0.3σ .

statistical argument for keeping the search program narrow. The marginal cost of scanning one more spectrum is small, and the cost of adding models to an existing one is exactly zero.

Uncountability is expensive. The argument above works only because N_{trials} can be counted. The moment the number of effective looks becomes a matter of opinion, which is the natural state of a neural-network-driven scan, the correction one has to defend is no longer $2 \ln N_{\text{trials}}$ with N_{trials} known but whatever a reviewer will accept. Section 6 quantifies how much that is worth paying to avoid.

3 The combinatorial scan, and what has never been looked at

Section 2 priced the program that exists: the spectra published searches scan. A scan does not have to be restricted to those. If an analysis builds every invariant mass it can from the objects in an event, the relevant budget is a different and much larger one, and the difference between the two is precisely the space nobody has examined.

3.1 Budget of a fully combinatorial scan

To keep this concrete rather than hypothetical we fix one object budget and price it. It is the shape a data-directed scan [1] of this kind naturally takes, and it is a design choice of this note rather than a description of any particular implementation:

- Categories are **exclusive** multiplicity bins in $(n_e, n_\mu, n_j, n_b, n_Z, \text{MET})$, so different categories select disjoint event sets and are statistically independent looks.
- The object budget is $n_e \leq 2, n_\mu \leq 2, n_j \leq 3, n_b \leq 3, n_Z \leq 1$, with $2 \leq \sum n \leq 4$, missing energy excluded from the masses.
- A single-lepton trigger requires each populated category to contain a lepton, either a loose e/μ or a leptonic Z .
- Missing energy splits every category in two but never enters a mass, and two same-flavour loose leptons split into opposite- and same-sign.
- Every size-2 to size-4 subset of the indexed objects is its own histogram, so $j_0 j_1, j_0 j_2$ and $j_1 j_2$ are three spectra rather than one.

This yields **150 exclusive multiplicity categories**, which become 204 once the same-flavour dilepton categories are split into opposite- and same-sign, and **1094 (category, mass-group) combinations**, or 1502 histograms after that same charge split. They span 82 distinct flavour compositions. Assigning each spectrum a resolution from its object composition ($\sigma/p_T = 0.04$ for $e/\mu/Z$, 0.10 for light jets and 0.20 for b -jets, combined as $r = \frac{1}{2} \sqrt{\langle \sigma_i^2 \rangle}$, which reproduces the per-channel values used in Section 2) and applying Eq. (2) gives

$$N_{\text{trials}} = 1.43 \times 10^5 \quad \implies \quad Z_{\text{local}} = 6.98 \quad (\text{band } 6.88\text{--}7.08),$$

so a local 5σ in such a scan is worth only 1.1σ globally. Relative to the published-program budget the trials factor grows by a factor 39 while the discovery bar moves by 0.5σ , which is the logarithm doing its work again. A combinatorial scan is not statistically prohibitive. It is about half a sigma more expensive than the program ATLAS already runs.

3.2 Which compositions has ATLAS actually scanned?

Those same 82 compositions give a natural denominator for a coverage question that is normally asked qualitatively. We map them onto the published bump observables of Appendix A generously,

counting a composition as covered if *any* published ATLAS search scans a mass built from those object types. A published “ V ” is allowed to be a Z , a jj pair or an $\ell\ell$ pair, a published “ t ” to contain b -jets, and so on. Even on that permissive reading:

Table 4: Reachable (≤ 4 -object) mass compositions with and without a published ATLAS bump hunt. The mapping is deliberately generous, so the uncovered column is a lower bound on the gap.

multiplicity	compositions	with a published scan	with none
$K = 2$	14	14	0
$K = 3$	28	7	21
$K = 4$	40	4	36
total	82	25	57

In this basis the gap is a pure multiplicity effect. Two-body space is saturated, while 75 % of three-body and 90 % of four-body compositions have never been scanned. An independent audit built from the publication record upward, in a different object basis {light jet, b -jet, lepton, photon}, reproduces the shape: 90 % of two-body, 25 % of three-body and roughly 11 % of four-body combinations scanned.

3.3 Three reasons the covered fraction is overstated

The generous mapping hides structure that sharpens the conclusion considerably.

Tag versus scan. Several searches that look like higher-multiplicity coverage in fact scan a two-body mass, with the extra object used only as a trigger or flavour tag. The b -tagged dijet search [5] is explicitly a two-body scan, and the dijet+lepton search [6] scans m_{jj} only. Counting either as multi-body coverage overstates the breadth of the record.

Almost all mixed-flavour coverage is three papers, all since 2022. Of roughly 45 distinct spectra ever bump-hunted by ATLAS, only about 15 are mixed-flavour or have three or more bodies, and essentially all of those come from [2, 7, 8]. One of the three is a machine-learning anomaly search; the other two are conventional scans that were simply never run before. Before 2022 that coverage was close to zero.

Most covered two-body spectra are covered only inside anomaly-selected phase space.

The broadest mixed-flavour scan in the ATLAS record [2] covers m_{jb} , m_{je} , m_{be} , $m_{j\mu}$, $m_{b\mu}$, $m_{j\gamma}$ and $m_{b\gamma}$ from 0.3 to 8 TeV, but does so inside autoencoder-selected regions rather than inclusive selections. There is no inclusive dedicated search for $m(\ell b)$, $m(\gamma b)$, or non-leptoquark $m(\ell j)$.

3.4 The programme that was meant to cover the rest

The ATLAS general search [3] populates 704 event classes, analyses 686 of them and defines more than 10^5 regions. It reads like blanket coverage. It is not, for a reason that bears directly on Section 2: it scans exactly two event-level observables per class, m_{eff} and the invariant mass of *all* visible objects. Class “ $1e2j$ ” therefore yields $m_{\text{inv}}(e, j, j)$ and m_{eff} , never $m(ej)$ or $m(jj)$. The general search is structurally blind to sub-combination masses, which is exactly where resonances live.

The paper states the reason outright: the observable set was restricted “to avoid a large increase in the number of hypothesis tests”. **ATLAS declined this coverage explicitly, on look-elsewhere grounds.** That is a deliberate trade rather than an oversight, and it is precisely the trade the numbers above reprice. Moving from two event-level observables to all

sub-combinations shifts the discovery bar from 6.4σ to 7.0σ . That is the entire cost of the coverage the general search gave up.

Two further points on that programme. It ran on 3.2fb^{-1} , which is 2015 data and 2.3 % of Run 2, and it has had no successor in the event-class format. The full-Run-2 anomaly-detection scan [2] is a successor in spirit, but it covers nine two-body spectra rather than hundreds of exclusive classes.

3.5 Named gaps

Concrete negatives, each checked individually against the publication record rather than inferred from the table above:

- $m(\ell\gamma)$, the only uncovered two-body composition in the basis. Excited leptons $\ell^* \rightarrow \ell\gamma$ were searched for at 7 TeV [9] and 8 TeV [10]; every 13 TeV excited-lepton search uses other channels.
- **Same-sign dilepton plus jets mass spectra.** $H^{\pm\pm} \rightarrow \ell^{\pm}\ell^{\pm}$ was a genuine same-sign mass scan at 7 TeV [11], but the full-Run-2 update [12] replaced the scan with a counting signal region.
- $m(jj\gamma)$, $m(bbj)$, $m(\ell\ell j)$, clean three-body negatives.
- $m(jjjj)$: no light-jet four-body scan. The b -tagged analogue *is* covered, by the resolved $X \rightarrow HH \rightarrow 4b$ channel [13].
- **Colour-sextet and E_6 diquarks**, $b^* \rightarrow b\gamma$ (CMS sets excited-bottom-quark limits in its γ +jet resonance search [14]; the ATLAS γ +jet searches are flavour-inclusive), $\tau^* \rightarrow \tau\gamma$ (the τ^* model was searched through contact interactions [15], so the $\tau\gamma$ channel is the gap), $m(t\gamma)$, and **same-sign $t\bar{t}$ as a mass bump** [16]: model or channel gaps with no ATLAS scan in any run.
- **The entire mixed-flavour $K = 3$ and $K = 4$ block:** $m_{jb\mu}$, m_{ejj} , $m_{e\mu b}$, m_{eejb} , $m_{\mu\mu bb}$ and the rest.

A fourth category turned up that is arguably more interesting than any of these, and is not “never examined”: signatures examined once at Run-1 luminosity and then quietly abandoned. $m(\ell Z)$ [17] and $m(Zb)$ [18] were both scanned at 8 TeV and never repeated. Re-running an abandoned 8 TeV scan on the full Run-2 dataset costs nothing in trials that Section 2 has not already counted.

A closing caution on how to read all of this. “Never scanned” is a statement about the publication record, not about whether the signal models exist or whether samples were ever generated for them. Those are orthogonal axes: a channel can be well supplied with theory and simulation and still have no published scan, which is precisely the situation the table above counts.

4 Does the arithmetic reproduce the excesses we have seen?

A trials-factor calculation is easy to get wrong in a way that is invisible. Nothing in Section 2 would look different if N_{trials} were off by an order of magnitude. There is one external check available, though, and it is a strong one. If the number of independent looks across the ATLAS search program is known even approximately, background-only statistics predicts how many 3σ and 5σ local excesses should have appeared. That prediction can be compared to the historical record.

4.1 The prediction

The relevant count is quasi-independent statistical tests, meaning mass-resolution elements plus signal-region bins, across all ATLAS searches rather than bump hunts alone.

Table 5: Trials-factor anchors for the full ATLAS search program.

level	looks N	basis
bump-hunt budget (public channels)	$\sim 3.7 \times 10^3$	this study, Section 2
two-body invariant-mass searches	$\sim 2\text{--}3 \times 10^4$	$> 50\,000$ LHC-wide [19], ATLAS $\sim$ half
full ATLAS BSM program (incl. SRs)	$\sim 5 \times 10^4$	300–500 searches $\times \mathcal{O}(100)$ regions [20]

With a one-sided Gaussian tail, $p(\geq 3\sigma) = 1.35 \times 10^{-3}$ and $p(\geq 5\sigma) = 2.87 \times 10^{-7}$, the expected counts are $N \cdot p$:

Table 6: Expected upward fluctuations under background-only.

N	expected $\geq 3\sigma$	expected $\geq 5\sigma$ (spurious)
10^4	14	0.003
5×10^4 (central)	67	0.014
10^5	135	0.029
resonance subset, 3.7×10^3	5.0	1.1×10^{-3}

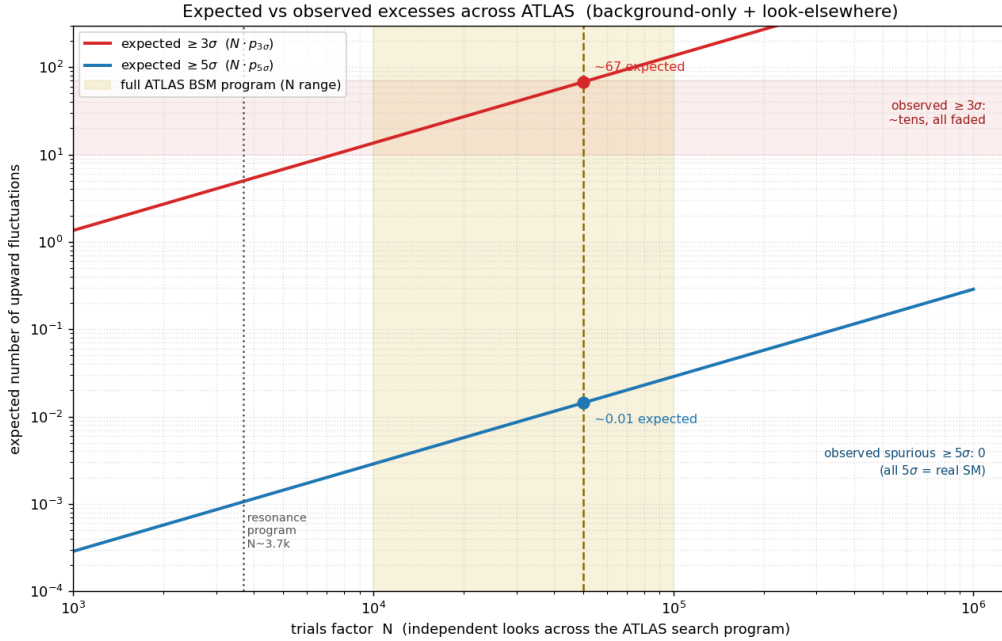


Figure 4: Expected number of upward fluctuations above Z as a function of the number of independent looks, with the anchors of Table 5 marked. The $\geq 5\sigma$ curve stays below unity across the entire defensible range of N , which is the quantitative content of the 5σ convention [21].

4.2 The record

At 3σ : order tens, essentially all faded. The 750 GeV diphoton excess (3.9σ local in 2015, 2.1σ global, gone with more data), the 2 TeV diboson WZ excess at 8 TeV, the 95 GeV and 151 GeV diphoton hints, the multilepton anomalies, a 400 GeV $t\bar{t}$ excess, and the surviving $\sim 3\sigma$

excesses found hiding inside published SUSY signal regions [20]. That is the expected population, not a mystery.

At 5σ : every one is a genuine Standard-Model process. The Higgs boson, four-top production, $t\bar{t}Z/t\bar{t}W/t\bar{t}H$, vector-boson scattering, single top; the public summary of ATLAS search results [22] contains no BSM counterexample. **Spurious BSM $\geq 5\sigma$: exactly zero.** Four-top production is instructive rather than a counterexample: it evolved from 3σ to 5σ and turned out to be real.

4.3 Verdict

Table 7: Expected versus observed, for $N \sim 5 \times 10^4$.

	expected	observed
$\geq 3\sigma$ local	$\sim 15\text{--}135$	order tens, all faded
$\geq 5\sigma$ spurious	~ 0.01	0

The two agree. This matters for the note’s central claim in a specific way: the same look-elsewhere arithmetic that produces $Z_{\text{local}} \approx 6.4$ in Section 2 also post-dicts the observed excess population without adjustment. The framework is at least self-consistent, and the parade of 3σ hints that come and go needs no explanation beyond $N \times p$.

The corollary is uncomfortable. With more than 50 000 looks LHC-wide, a local 5σ in a fully agnostic scan is worth only 1.5 to 2.5σ globally, which is the basis of the argument [19] for raising the agnostic-scan threshold towards 7σ . This note reaches essentially the same number from the opposite direction for a combinatorial scan (Section 3, $Z_{\text{local}} = 6.98$).

Three caveats. N spans a factor ten and is the dominant uncertainty here, although the conclusion survives the whole range. The counts mix ATLAS-only and ATLAS+CMS excesses. And “order tens observed” counts *notable* excesses; the raw number of individual $\geq 3\sigma$ bins is much larger and goes unremarked, which is also consistent, since those are the expected noise.

5 Selecting candidates from a scan

Sections 2 and 3 priced the scan. This section asks what to do with its output. A broad scan produces a large number of local significances, one per bin per spectrum, and something has to decide which of them is worth following up. Three rules are compared here on identical footing: take the largest (*argmax*), take everything above a fixed threshold, or let the data set the cut through a false-discovery-rate criterion.

The setting is deliberately minimal: $n = 30\,000$ bins, of which $n - 1$ are background with $z_i \sim \mathcal{N}(0, 1)$, and at most one contains a signal with $z \sim \mathcal{N}(\mu, 1)$. Whatever passes stage 1 is unblinded in an independent sample B and confirmed if $z_B > 3$. This is the two-stage structure of Section 6 stripped to its statistical core.

5.1 The background maximum is not Gaussian

The first thing to get right is the null distribution of $\max_i z_i$. Two limits are routinely confused. Running more pseudo-experiments makes the histogram converge to the *exact* density

$$f_M(x) = n \phi(x) \Phi(x)^{n-1}, \quad F_M(x) = \Phi(x)^n, \quad x_q = \Phi^{-1}(q^{1/n}), \quad (3)$$

which is exact for every n . Increasing n makes the *rescaled* f_M converge to a Gumbel law [23, 24], and that convergence is only logarithmic, with relative error $\mathcal{O}(1/2 \ln n)$.

At $n = 30\,000$ the difference is measurable, and 10^4 pseudo-experiments are enough to resolve it (Table 8). The exact form is confirmed; the asymptotic Gumbel is rejected at $p = 2 \times 10^{-15}$ and a best-fit Gaussian at $p = 6 \times 10^{-30}$.

Table 8: Distribution of the maximum of $n = 30\,000$ standard normals, from 10^4 pseudo-experiments, against the exact law and its Gumbel limit ($a_n = 4.005$, $b_n = 0.220$).

	simulated	Gumbel asymptotic
mean	4.110	4.132
standard deviation	0.284	0.282
median	4.072	4.086
skewness	0.89	1.14
KS vs exact $\Phi(x)^n$	$D = 0.0077$, $p = 0.59$	✓
KS vs Gumbel(a_n, b_n)	$D = 0.041$, $p = 2 \times 10^{-15}$	
KS vs best-fit Gaussian	$D = 0.058$, $p = 6 \times 10^{-30}$	

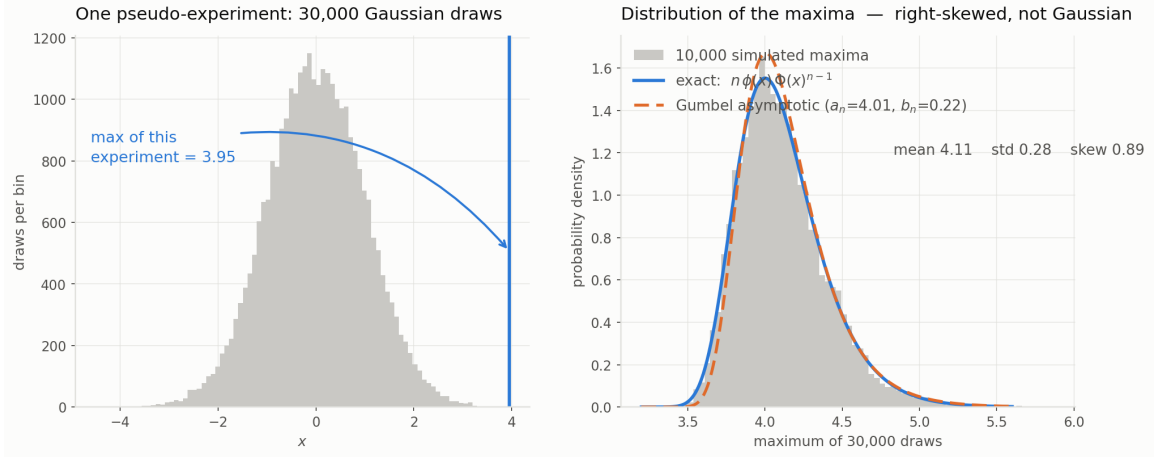


Figure 5: Left: one pseudo-experiment of 30 000 Gaussian draws, with its maximum marked. Right: the distribution of 10^4 such maxima, with the exact density of Eq. (3) and the asymptotic Gumbel overlaid. The distribution is strongly right-skewed, so a symmetric error bar on a maximum is never appropriate.

The practical rules that follow are short. Gumbel is fine for intuition. For any actual number, whether a threshold, a p -value or a quantile, use $F_M = \Phi^n$, which costs nothing and is right. Never model a maximum as Gaussian. This also connects back to Section 2: the familiar trials factor is the leading term of $1 - \Phi(z)^{N_{\text{eff}}}$, and in the far tail, where global p -values live, a few-percent Gumbel error is not something one wants between oneself and a discovery claim.

5.2 A real signal often loses the scan

The probability that the signal bin *is* the maximum, $p(\mu) = \int \phi(s - \mu) \Phi(s)^{n-1} ds$, is sobering. A genuine 4σ signal wins the scan only 46 % of the time, because it is racing a distribution centred at 4.11 with a spread of 0.28. The two-stage confirmation probability follows as $p_{\text{conf}} = p(\mu) \Phi(\mu - 3) + [1 - p(\mu)] \Phi(-3)$ (Table 9).

The second stage is nearly free. For $\mu \geq 5\sigma$ the loss between “wins the scan” and “confirmed” is under two percentage points, because a signal strong enough to win stage 1 almost always exceeds 3σ in stage 2. **Stage 2 also carries no trials factor**, since the bin was chosen by stage 1. The false-alarm rate per opening is $1 - \Phi(3) = 1.35 \times 10^{-3}$, or 3.0σ *global*, whatever the size of the scan that selected it.

5.3 A threshold dominates the argmax

Taking the maximum throws information away: it ranks bins and then discards everything about their absolute size. The alternative is to select every bin above a fixed threshold t . Under background-only the number selected is Poisson with mean $\lambda(t) = (n-1)[1 - \Phi(t)]$, so the natural operating point is $t^* = \Phi^{-1}(1 - 1/(n-1)) = 3.99$, where $\lambda = 1$. That is exactly the argmax’s fake budget of one candidate per experiment, and at that matched budget the threshold rule wins at every signal strength (Table 9).

Table 9: Probability that a signal of strength μ is confirmed in sample B, for the three stage-1 rules, all at the same background-only budget of one candidate per experiment (3.0σ global false-alarm rate). The threshold operating point is $t^* = 3.99$ and the matched false-discovery rate is $q^* = 0.381$.

μ	wins the scan	argmax confirmed	BH at q^*	threshold at t^*
3σ	14.2 %	7.2 %	7.5 %	8.1 %
4σ	45.8 %	38.6 %	40.3 %	42.5 %
5σ	80.3 %	78.5 %	80.6 %	82.5 %
6σ	96.4 %	96.3 %	97.2 %	97.7 %

Putting both rules on a common ROC, with the average number of false confirmations on one axis and the signal confirmation probability on the other, makes the structure clear (Figure 6). The threshold rule traces a curve as t varies. The argmax is a *single point*, because it always selects exactly one bin, and that point lies strictly inside the curve at every μ . It is dominated, not merely different.

One caveat took some care to get right. The bare ratio of power to false positives has no interior maximum; it increases monotonically with t and simply says “cut harder”. The well-posed question is where the ROC tangent has a given slope, which amounts to saying what a false positive costs relative to a discovery. Once that cost is specified, the optimum has a closed form.

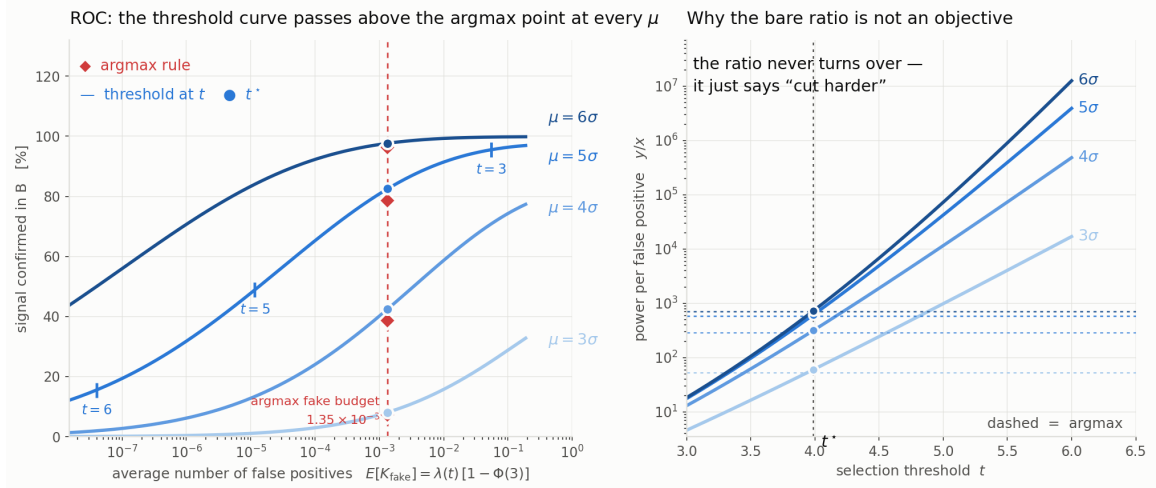


Figure 6: Left: the ROC of the threshold rule (one curve per signal strength) with the argmax rule as a single point (diamonds), at $\mu = 3, 4, 5, 6\sigma$. The argmax point lies inside the curve at every μ , so at the same false-positive budget a threshold confirms more signal. Right: the bare power-per-false-positive ratio, which never turns over. An objective has to specify what a false positive costs.

5.4 Letting the data set the cut: Benjamini–Hochberg

The third rule replaces a fixed bar with a false-discovery-rate criterion [25]. Convert each bin to a one-sided p -value, sort, and reject the $K(q)$ smallest, where $K(q) = \max\{k : p_{(k)} \leq kq/n\}$. The dial is now q , the nominal FDR. Since K is a random variable depending on the whole p -vector, this has to be evaluated by Monte Carlo.

Two properties matter here. First, under the global null the probability that BH makes at least one rejection is exactly q . This follows from the identity $P(\sup_t F_n(t)/t \geq 1/q) = q$ for the uniform empirical distribution, usually attributed to Daniels [26], and our Monte Carlo reproduces it ($q = 0.05 \rightarrow 0.0505$, $0.2 \rightarrow 0.1990$, $0.5 \rightarrow 0.5008$). The *number* of rejections is not q : BH is bursty, and past q^* multi-rejection events dominate. Second, the realised cut is not fixed. In a pseudo-experiment where BH rejects K bins the weakest accepted bin sits at $z_{\text{cut}} = \Phi^{-1}(1 - p_{(K)})$, which fluctuates from experiment to experiment.

Matching BH to the same fake budget as the other two rules, $E[K | H_0] = 1$, gives $q^* = 0.381$, and it then agrees on both false-alarm measures (1.34×10^{-3} expected false confirmations and $P(\geq 1) = 1.336 \times 10^{-3}$, which is 3.0σ global). Its effective single-bin bar is $z_1(q^*) = 4.21$, *stricter* than $t^* = 3.99$.

BH lands strictly between the other two rules (Table 9, Figure 7). It beats the argmax for the same reason the threshold does: it never discards the signal candidate merely because some background bin happened to be larger, and the step-up lets a signal ride along at rank 2 or below. It loses to the fixed threshold because keeping $E[K | H_0] = 1$ while spending part of that budget on rank-2+ clumps forces the single-bin bar up to 4.21, and those extra selections are pure waste. They are background bins, and the signal, normally rank 1, gains nothing from them. The effect is specific to large q ; at small q BH is essentially the rank-1 bar and the three rules converge.

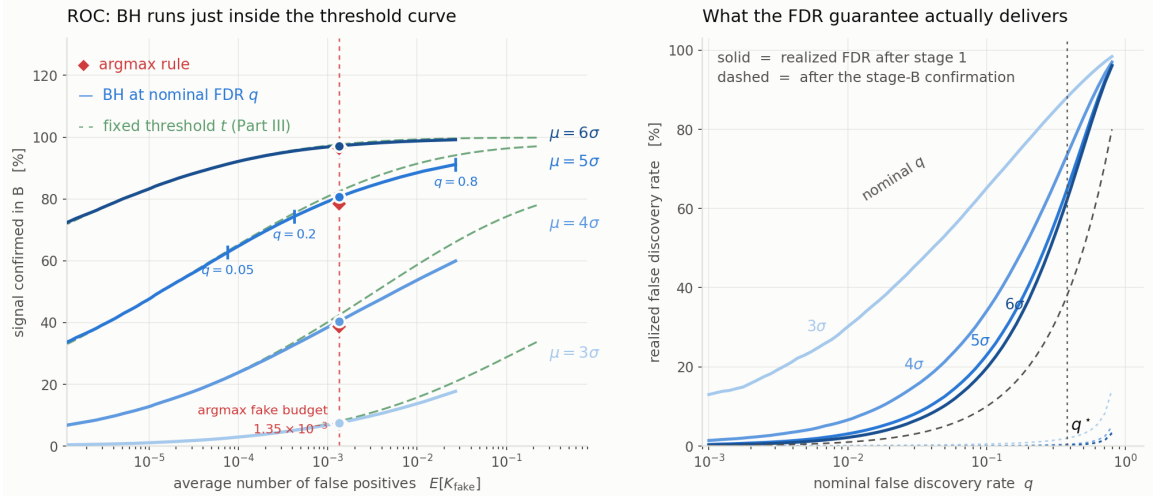


Figure 7: Left: all three stage-1 rules on the ROC of Figure 6. BH runs just inside the fixed-threshold curve and outside the argmax point. Right: the realised false discovery rate before and after the stage-B confirmation. The confirmation step drives it to near zero regardless of the nominal q .

5.5 The same three rules with an imperfect estimator

Everything above assumes the estimator returns $z \sim \mathcal{N}(0, 1)$ for every background bin. Section 6.7 argues that a network-driven scan does not have that estimator, and the comparison is worth redoing with the same defect model, because the three rules are not equally exposed to it. The exposure is structural rather than numerical: argmax and BH are **rank-based**, so an artefact anywhere in the scan changes the decision about every other bin, while a fixed threshold is **absolute** and an artefact at 1 TeV says nothing about the bin at 3 TeV.

Table 10 redoes Table 9 at $\mu = 5\sigma$, with every rule tuned to the *same* 3.0σ -global false-alarm budget under each contamination. The $\epsilon = 0$ row reproduces the perfect-estimator result, which fixes the comparison.

Table 10: Probability that a 5σ signal is confirmed, for the three stage-1 rules, all held at 1.35×10^{-3} expected false confirmations. GLITCH and BIAS are the incoherent and coherent defect classes of Section 6.7. “n/a” means the rule cannot be tuned to that budget at all. Monte-Carlo error on the BH column is ± 0.35 points; the other two are analytic.

estimator	argmax	threshold	BH	thr – argmax	thr – BH
perfect (Table 9)	78.5 %	82.5 %	80.9 %	+4.0	+1.6
GLITCH $\epsilon = 10^{-4}$	73.2 %	81.2 %	80.1 %	+8.0	+1.1
GLITCH $\epsilon = 10^{-3}$	43.8 %	57.3 %	57.2 %	+13.6	+0.1
GLITCH $\epsilon = 10^{-2}$	6.2 %	5.8 %	6.0 %	−0.4	−0.2
BIAS, any ϵ	<i>untunable</i>	$\sim 0\%$	<i>unreachable</i>	—	—

The threshold’s margin over the argmax widens, from +4 to +13.6 points. The cause is visible in a single diagnostic: the probability that the argmax *is* the signal falls from 80.3 % to 74.9 % to 44.8 % as ϵ runs from 0 to 10^{-3} . At $\epsilon = 10^{-3}$ some thirty artefact bins are scattered through a 30 000-bin scan, and any one of them outranking the signal removes it completely. A rank-1 rule has no graceful degradation, whereas a threshold pays only through the extra candidates it forwards.

Contamination strips BH of the adaptivity that made it attractive. BH converges onto the fixed threshold ($+1.6 \rightarrow +1.1 \rightarrow +0.1$), and the reason is mechanical: to hold the false-alarm budget it has to be run at ever smaller q , and BH at small q simply *is* a threshold at $\Phi^{-1}(1 - q/n)$. One keeps the threshold’s performance while losing the ability to interpret the dial.

And that dial becomes actively misleading. The nominal q needed to buy the same 3.0σ -global rate runs $0.38 \rightarrow 0.32 \rightarrow 1.4 \times 10^{-2} \rightarrow 4.2 \times 10^{-7}$ across the rows of Table 10 (Fig. 8, middle). Under coherent defects it cannot be bought at all: even $q = 10^{-9}$ leaves 2.5×10^{-3} expected false confirmations at $\epsilon = 10^{-4}$, rising to 0.25 at $\epsilon = 10^{-2}$, because a coherent artefact carries a p -value small enough to be rejected at essentially any q and then confirms in stage 2. Nothing here contradicts the FDR guarantee — BH controls the false discovery rate under the null it is handed, and a contaminated scan does not hand it that null.

The argmax should be dropped rather than deprioritised. It has no tuning parameter, so its false-alarm rate is whatever the contamination makes it: $1.35 \times 10^{-3} \rightarrow 0.137 \rightarrow 0.687 \rightarrow 0.957$ under coherent defects (Fig. 8, right). At $\epsilon = 10^{-2}$ the argmax rule produces a false confirmation in 96 % of scans and there is no way to dial it back. Under incoherent defects it stays at 1.35×10^{-3} , since a glitch dies in stage 2, and only the power collapses.

The last row of Table 10 is the same result as Section 6.7 seen from a different direction: a coherent defect is indistinguishable from a signal, so no selection rule can help against it. The threshold remains formally calibratable — it can always be raised, and its reachable floor is 10^{-9} against BH’s 2.5×10^{-3} — but only at essentially zero power. That is a background-modelling problem, not a selection-rule problem.

Two caveats. The bins here are independent, where a real scan has correlated neighbouring mass bins; and the defect magnitude is fixed at 2σ , so the size of the gaps scales with it while the ordering does not. At the heaviest incoherent contamination all three rules are equally dead and the argmax nominally edges ahead by a few tenths of a point, which is not a meaningful ranking.

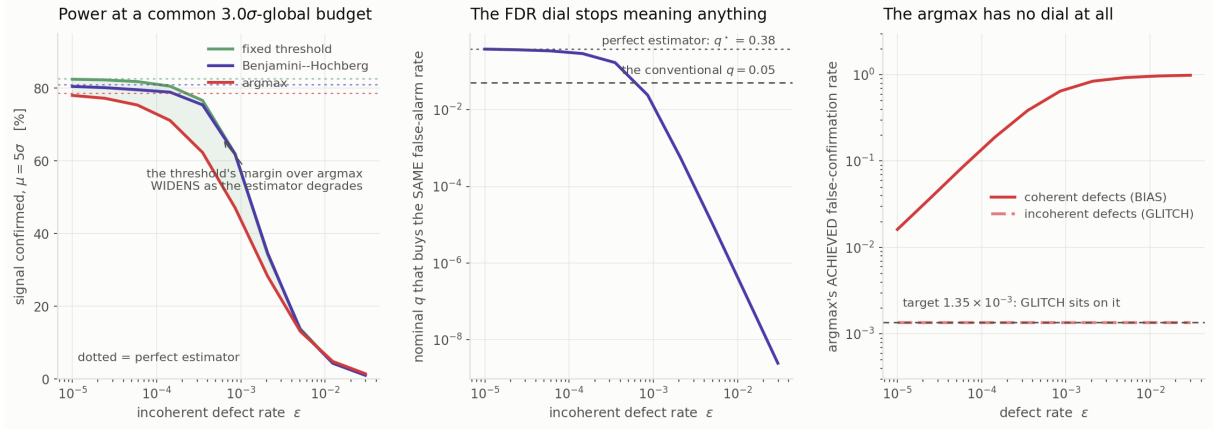


Figure 8: **Left:** confirmation probability for a 5σ signal as the estimator degrades, all three rules held at the same false-alarm budget; dotted lines are the perfect-estimator values. **Middle:** the nominal FDR that buys that budget, against the conventional $q = 0.05$. **Right:** the argmax’s achieved false-alarm rate, which it has no parameter to control.

5.6 What this implies for a data-directed scan

1. **Do not select by rank alone.** “Take the most significant bin” is a dominated strategy. Selecting on absolute significance recovers the information the ranking throws away, at identical false-alarm cost, and gains a few percentage points of power at every signal strength. With an imperfect estimator the argument gets stronger rather than weaker (Section 5.5): the margin grows to more than ten points, and the argmax loses control of its false-alarm rate entirely.
2. **Fix the fake budget, not the threshold.** The natural, scan-size-independent statement of an operating point is “one expected background candidate per experiment”, which determines t^* from n . Quoting a bare z threshold hides the scan size.
3. **An FDR rule is a defensible middle ground, but never quote the nominal q as a false-alarm specification.** BH costs a little power relative to the matched threshold, and it comes with a guarantee that is easy to state to a reviewer. What that guarantee does not survive is a miscalibrated tail: the q needed to buy a given false-alarm rate moves by six orders of magnitude across the contamination range of Section 5.5, and under coherent defects no q buys it. Whichever rule is adopted, calibrate and quote the *achieved* rate.
4. **The confirmation stage is cheap and carries no trials factor.** That is the structural fact the next section is built on.

6 Two-stage A/B unblinding

6.1 The proposal

Split the dataset into a fraction f (sample A, exploration) and $1 - f$ (sample B, confirmation). Scan *everything* in A with *no* global correction, since what happens in A is selection rather than inference. Pre-register every window with local $Z_A \geq Z_{\text{cut}}$. Unblind only those windows in B, and make the discovery claim on the B-only p -value, corrected for the k pre-registered windows.

The attraction is immediate. The trials factor in B is not modelled but *counted*: it is the number of lines in a frozen document. For a classical fixed-window search that is a modest convenience. For a neural-network-driven scan, where “how many independent looks did the network take” has no crisp answer, it is the difference between a defensible number and an argument.

6.2 The cost

Nothing is free, and here the trade is explicit. One avoids a logarithmic penalty by giving up luminosity, which costs $\sqrt{f}$ on the significance. For a signal whose full-dataset local significance would be Z_{full} , the two stages fluctuate independently, so the joint discovery probability is

$$P_{\text{disc}} = \Phi\left(\sqrt{f} Z_{\text{full}} - Z_{\text{cut}}\right) \cdot \Phi\left(\sqrt{1-f} Z_{\text{full}} - Z_B^{\text{req}}\right), \quad Z_B^{\text{req}} = \sqrt{25 + 2 \ln k_{\text{eff}}}, \quad (4)$$

with $k_{\text{eff}} = N p_1(Z_{\text{cut}}) + 1$ the expected number of pre-registered windows. The reach is the Z_{full} at which $P_{\text{disc}} = 50\%$.

The distinction between joint power and the naive median arithmetic matters. Requiring each stage to succeed at its median gives $Z_{\text{med}} = \max(Z_{\text{cut}}/\sqrt{f}, Z_B^{\text{req}}/\sqrt{1-f})$, which *underestimates* the required signal wherever both constraints bind, since two independent coin flips at 50% each give 25% jointly. Equation (4) is the honest figure of merit, and toy Monte Carlo reproduces it to 0.03σ .

At this study's budget ($N = 6616$, single-stage reach 6.53σ):

Table 11: Signal strength required for a 50% probability of discovery, as a function of the splitting scheme, at $N = 6616$ independent looks.

procedure	reach Z_{full}	cost
single-stage full scan, exactly corrected	6.53	—
optimised split ($f \approx 0.2$ – 0.3 , $Z_{\text{cut}} \approx 2$ – 3)	~ 7.0	$+0.5\sigma$
naive 50/50 split, $Z_{\text{cut}} = 3$	~ 8.0	$+1.4\sigma$

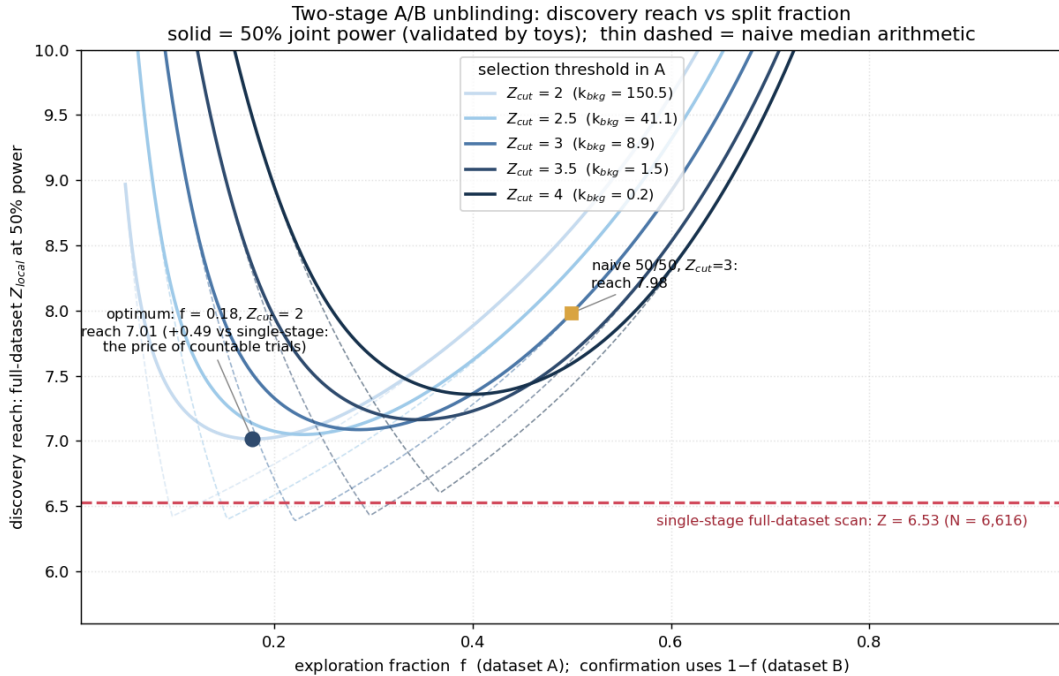


Figure 9: Discovery reach versus exploration fraction f , for several stage-1 selection thresholds. Solid curves are 50% joint power (Eq. (4)), thin dashed the naive median arithmetic. The red line is the exactly corrected single-stage scan. The optimum is markedly asymmetric, with a small exploration sample and a large confirmation sample, and the naive 50/50 split is a poor choice.

6.3 The look-elsewhere effect is conserved, not reduced

It is tempting to expect a trials count large enough to make splitting pay. There is none, and the reason is an identity rather than a numerical accident. The cost $2 \ln N$ factorises:

$$2 \ln N = 2 \ln \frac{N}{k} + 2 \ln k,$$

where the first term is paid implicitly, as the stage-1 selection cut that reduces N looks to k pre-registered windows, and the second explicitly in B. Splitting *relocates* the penalty rather than removing it. What remains is the two-coin penalty, the requirement that both independent halves succeed, which runs from $+0.3$ to $+0.5\sigma$ between $N = 10$ and $N = 10^{10}$ and never crosses zero.

The number of unblinded windows barely matters: moving k from 1 to 1000 changes the reach by 0.15σ , so k can be chosen on practical grounds ($k \approx 10$).

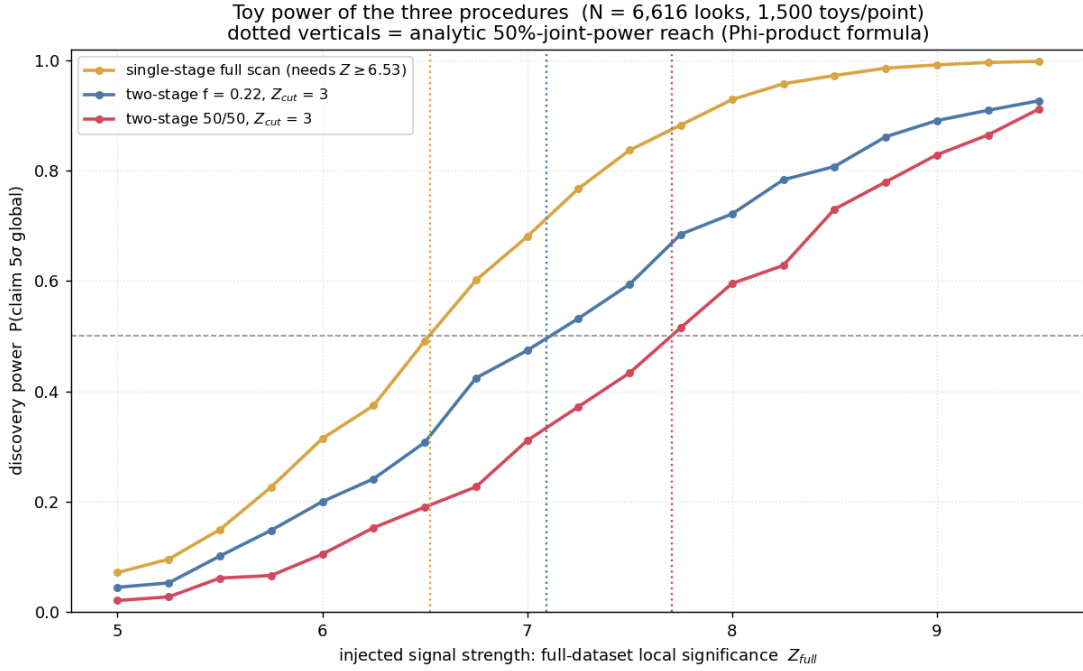


Figure 10: Toy Monte-Carlo validation: discovery power versus injected signal strength for the three procedures, with the analytic median-reach values as vertical lines. The toys treat the budget’s N effective looks as independent standard normals under background-only and recover the full-dataset scan exactly as $z_{\text{full}} = \sqrt{f}z_A + \sqrt{1-f}z_B$.

6.4 Would swapping the halves recover it?

The natural objection to that identity is that half the luminosity sits idle at each stage. Why not run *both* directions — explore in A and confirm in B, *and* explore in B and confirm in A — and claim if either confirms? This is two-fold cross-validation of the *claim*, and it is a genuinely different procedure rather than a relabelling.

The trade is explicit. Both directions pre-register, so the confirmation windows double, $k \rightarrow 2k$, costing $2 \ln 2$ on the B bar: at $Z_{\text{cut}} = 3$ and $N = 6616$, Z_B^{req} moves from 5.64 to 5.75. What is bought is a second chance at the same signal. Since $Z_B^{\text{req}} > Z_{\text{cut}}$, the two directions coincide only where *both* halves clear the claim bar, so the union power is

$$P_{\text{disc}}^{\text{sym}} = \Phi_{\text{cut}}^A \Phi_{\text{req}}^B + \Phi_{\text{cut}}^B \Phi_{\text{req}}^A - \Phi_{\text{req}}^A \Phi_{\text{req}}^B, \quad \Phi_x^A = \Phi\left(\sqrt{f} Z_{\text{full}} - Z_x\right), \quad (5)$$

and Φ_x^B the same with $f \rightarrow 1 - f$; toys reproduce it to 0.01σ . At fixed thresholds the gain over one direction is a factor $2 - \Phi_{\text{req}}/\Phi_{\text{cut}}$, that is, at most two.

That gain is collected only when both directions have comparable power, which means $f = 1/2$ — and $f = 1/2$ is precisely the split the one-way analysis rejects, because confirmation is the luminosity-hungry stage. The swap therefore does not beat the asymmetric scheme; it *rescues the symmetric one*.

Table 12: One-way splitting against the symmetrised swap at $N = 6616$ (single-stage reach 6.53). The first two rows re-optimize f at fixed Z_{cut} ; the last optimises (f, Z_{cut}) inside the practical design box $Z_{\text{cut}} = 2\text{--}4.5$.

design	one-way		swapped	
	reach	cost	reach	cost
50/50, $Z_{\text{cut}} = 3$	7.98	+1.45	7.38	+0.86
$f = 0.30$, $Z_{\text{cut}} = 3$ (recommended)	7.09	+0.56	7.15	+0.63
best design in the practical box	7.01	+0.49	7.03	+0.51

Applied on top of the recommended working point the swap is actively harmful: the second direction would have to confirm a 5σ -global result on 30% of the data, so it almost never fires, and one pays the $2\ln 2$ for a dead direction. Optimised freely inside the practical box the two schemes are a wash, $+0.49\sigma$ against $+0.51\sigma$, and the gap stays within 0.11σ over nine decades of N , changing sign with the design box rather than crossing over at any particular trials count.

Why neither can win is clearest geometrically. In the (z_A, z_B) plane the Neyman–Pearson acceptance region for a mean shift is the half-plane $\sqrt{f} z_A + \sqrt{1-f} z_B > \sqrt{25 + 2\ln N}$ — which is exactly the single-stage scan on the recombined dataset. One-way splitting approximates that half-plane with an L -shaped corner, the swap with a two-step staircase. More steps track the line more closely, which is why the swap repairs the 50/50 split, but every step multiplies the pre-registered windows. Conservation of the look-elsewhere effect is that staircase-versus-half-plane inefficiency, and no number of folds removes it.

The decisive objection, though, is not statistical. Of the three things the split buys (Section 6.5) the swap keeps two and destroys the third. The trials factor is still counted and still auditable — $2k$ is as countable as k — but procedural blindness is gone: after the first direction the analyst has seen B, so the second direction’s “confirmation” on A is made by someone who already knows both halves. It is valid only if the entire two-directional procedure is frozen and executed in a single automated pass, and a review committee is entitled to discount it even then. The swap also does nothing about the correlated-systematics mode of Section 6.6, which for a data-directed analysis is the leading one.

None of this argues against symmetrising the *training*, which is a separate question taken up in Section 6.7: holding the background model out of the sample it scores is nearly free and attacks the sculpting mode directly. Symmetrising the *claim* is the thing that costs, and on this budget it buys nothing.

6.5 When the split is nonetheless the right choice

The split wins if and only if the single-stage trials factor one would have to *defend* exceeds $R^* \approx 13$ times the true N . For a classical fixed-window scan that is implausible, and the corrected single stage wins outright. For a network-driven hunt a factor of 13 in forced conservatism is entirely imaginable, and that, rather than the logarithm, is the quantitative case for two-stage unblinding. It buys three things:

- an **exactly countable** trials factor in B,

- **auditability**: the correction is the number of lines in a document frozen before unblinding, and
- **procedural blindness**: the analysis cannot be tuned on the confirmation sample.

6.6 The failure mode splitting does *not* address

A/B splitting attacks the *statistical* look-elsewhere effect only. A spurious bump caused by correlated systematics, such as background mis-modelling or a network that has learnt to sculpt a peak, appears identically in both halves and will happily “confirm”. For a data-directed analysis this is not a corner case but the leading failure mode. The split protects against fluctuations, not against the analysis fooling itself. That statement is exact but incomplete: between a pure fluctuation and a fully correlated systematic lies everything an imperfect estimator does, and Section 6.7 prices the split over that range.

6.7 What changes if the significance estimator is imperfect

Everything above assumes a perfectly calibrated estimator: every look returns $Z \sim N(0,1)$ under background, so the only thing to control is the look-elsewhere effect, and splitting can only lose. A network-driven scan does not have that estimator. It has one that occasionally returns an outlier — a fit that failed to converge on this subsample, a background model that has absorbed a fluctuation, a window where the shape is simply wrong. Repricing the split against that estimator changes the verdict, and the reason it changes is worth more than the number.

The only property of an outlier that an A/B split can see is whether it *repeats in the other half*. That suggests three classes, distinguished by nothing but their correlation across the split:

Table 13: Classes of estimator defect, by how they correlate between the two halves. Only the last column matters to a split.

class	physical origin	behaviour across the split
BIAS	mismodelled background or detector artefact in that window	the pull is $\delta B/\sqrt{B} = \delta\sqrt{wB}$, so it grows with luminosity exactly like a signal and is identical in both halves
VAR	the look’s uncertainty is underestimated by a factor s	the miscalibration is shared, but what it amplifies is each half’s <i>independent</i> fluctuation
GLITCH	a per-run failure: non-convergence on this subsample, a network artefact, a corrupted input	redrawn independently every time the estimator is run

The mechanism is a one-line geometric statement, and Fig. 11 is the whole section in a picture. In the (Z_A, Z_B) plane a coherent pull b places a look at $(\sqrt{f}b, \sqrt{1-f}b)$ — on the locus $Z_B = \sqrt{(1-f)/f} Z_A$, which is precisely where a real signal of the same strength sits. An incoherent one places it near the $Z_B \approx 0$ axis. The split is a cut that separates the axis from the locus, and by construction cannot separate the locus from itself. It follows immediately that a coherent defect is not merely hard for the split to reject: it is *exactly as confirmable as a signal*, which is the content of Section 6.6 restated.

The same two failure modes on an actual falling spectrum are shown in Fig. 12: a background fit that undershoots by $\sim 10\%$ in the A half alone produces a 4.1σ excess that is duly pre-registered and then collapses to 2.6σ in B, while a background genuinely mismodelled by 8.8% gives 4.2σ in A and 6.9σ in B and clears the claim bar.

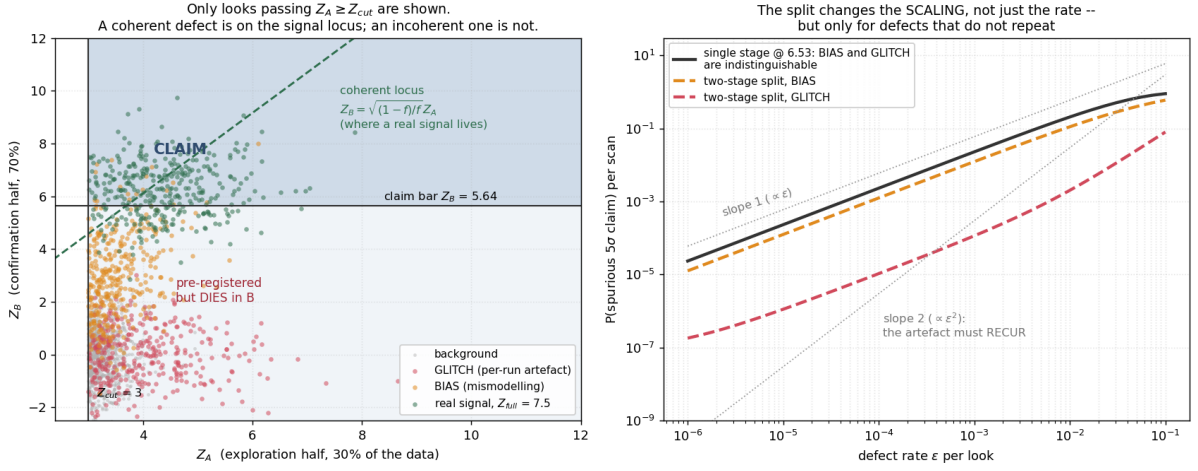


Figure 11: Left: looks that pass the stage-1 cut, in the plane of the two half-dataset significances. Real signals and coherent mismodellings populate the same locus; per-run artefacts sit on the $Z_B \approx 0$ axis and die at the claim bar. **Right:** the resulting spurious-claim rate. The two defects are *identical* to a single-stage scan — one curve, by construction — and the split separates them by changing the scaling from ϵ to ϵ^2 for the incoherent one alone.

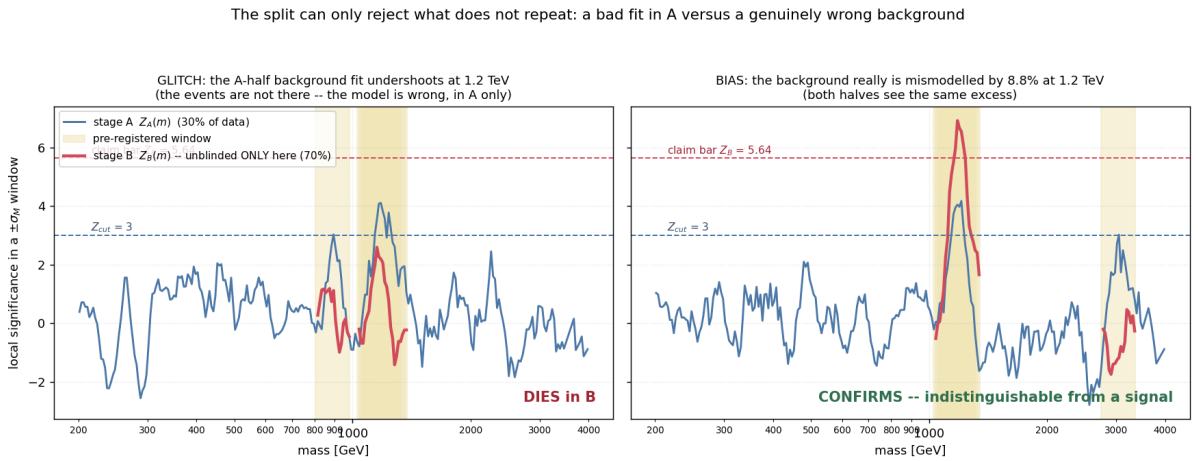


Figure 12: Toy spectra scored the way an analysis would score them, at the recommended working point. **Left:** the A-half background model is locally wrong, so the excess exists only in the estimator and stage B removes it. **Right:** the background model is wrong for both halves, so the excess is real in the data and confirms. Only the first of these is a fluctuation in the sense the split can act on.

The comparison cannot be made on raw false-alarm rates, because the single stage can always buy robustness by raising its bar. The honest figure of merit is therefore *matched*: raise the single-stage threshold until it is equally likely to make a spurious claim, and only then compare reach. Table 14 does this at the recommended working point, with each defect occurring at rate $\epsilon = 10^{-3}$ and magnitude $\sim 2\sigma$.

Table 14: Matched-robustness comparison at $f = 0.30$, $Z_{\text{cut}} = 3$, $N = 6616$, for a defect rate $\epsilon = 10^{-3}$ (≈ 1 artefact above 3σ per scan). A positive gain means the split reaches a *weaker* signal than a single-stage scan that is equally safe.

estimator	suppression	split reach	matched bar	gain
perfect (the assumption above)	—	7.09	6.53	-0.56σ
BIAS	$1.9\times$	7.08	6.95	-0.14σ
VAR ($s = 2$)	$3.5\times$	7.10	7.20	$+0.10\sigma$
GLITCH	$199\times$	7.11	9.61	$+2.50\sigma$

Against a per-run glitch the split suppresses the spurious-claim rate by two orders of magnitude, because the artefact must recur *independently* in B above 5.6: the rate goes as ϵ^2 rather than ϵ . A single-stage scan would have to demand $Z \geq 9.6$ to be as safe. Against a coherent bias the split suppresses by $1.9\times$, but that is not discrimination — it is merely the split’s own worse reach hurting fakes along with signal — and the single stage buys the same factor for $+0.42\sigma$, which is cheaper than the split costs. Section 6.6 is the $\rho = 1$ limit of this table.

Coherence is the deciding variable

Interpolating between the two extremes with a coherent fraction ρ (Fig. 13), the break-even sits at $\rho^* \approx 0.80$ and moves by less than 0.02 over two decades in the overall defect rate. The rule is therefore unusually portable:

If fewer than about four in five of the estimator’s outliers repeat in the other half, the split pays for itself.

Under a plausible cocktail — coherent bias, bad fits and glitches each at 10^{-3} , together ≈ 3 artefacts above 3σ per scan — the net is $+0.29\sigma$ in the split’s favour, against -0.56σ for a perfect estimator. And ρ is measurable *before* unblinding: split the sidebands or MC-only pseudo-data, and count what fraction of the 3σ excesses recurs.

Where the training arrangement enters

The sharpest case is a background model *fit to data*, which is the data-directed situation. Fitting it on the same events one then tests for excesses is overfitting, and overfitting is a GLITCH-type defect tied to the specific events in that half. A single pass cannot escape it: there is no held-out half. Three arrangements are available to a split, and they are not equivalent:

1. **self-contained** — train on A and score A, train on B and score B. Both stages overfit, but independently, so the claim is ϵ^2 -suppressed. Bookkeeping exact. Worth $+2.50\sigma$.
2. **full cross** — train on B and score A, train on A and score B. No overfitting anywhere, worth $+5.67\sigma$, *but* the stage-A selection is now a function of B through the trained model, so the pre-registration is no longer independent of the confirmation sample and the k -coin p -value is no longer exact.
3. **A-trained throughout** — train on A, score A to select, then apply that same frozen model to B. The confirmation Z is produced by an estimator that has never seen a single B event,

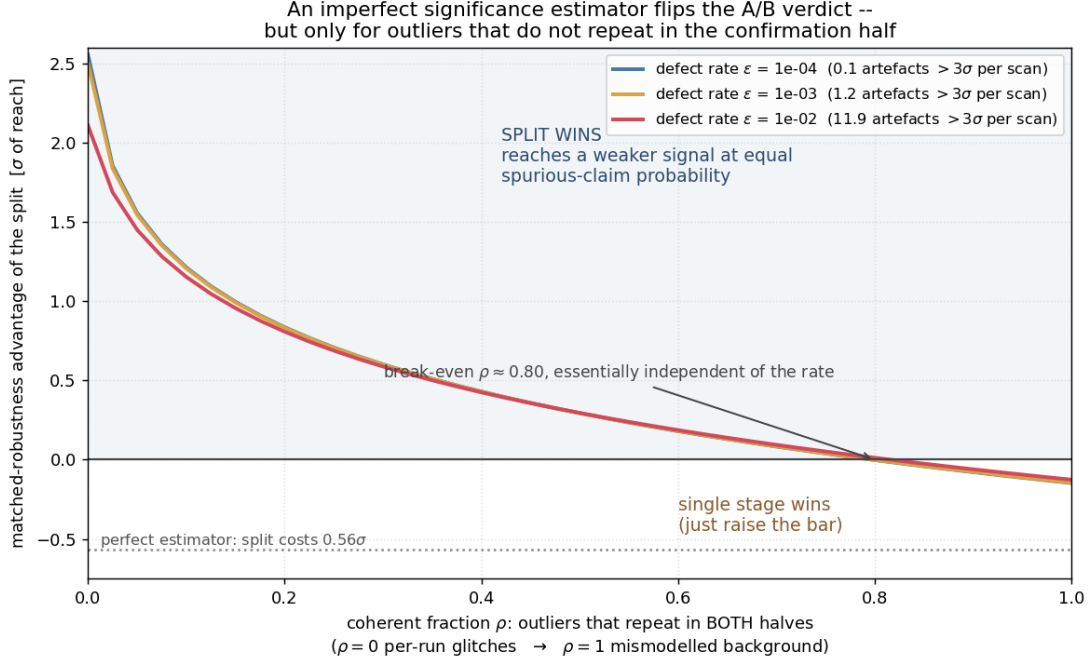


Figure 13: Matched-robustness advantage of the split as a function of the coherent fraction ρ of the estimator’s outliers, at three defect rates. The dotted line is the pure cost of splitting with a perfect estimator. The break-even is essentially rate-independent, so the criterion is a property of the estimator’s failure modes rather than of how often they occur.

so overfitting is removed *by construction* rather than beaten down statistically; and nothing about B enters the pre-registration, so the bookkeeping stays exact. Worth $+5.64\sigma$.

Arrangement 3 dominates: it recovers all but 0.03σ of the full-cross gain while keeping the blindness that made the scheme worth adopting. The reason it can is that the overfitting it leaves in place — in stage A — only inflates the candidate list (k : $8.9 \rightarrow 10.1$), and k enters the bar as $2 \ln k$. **Contaminating the exploration stage is logarithmically cheap; contaminating the confirmation stage is what costs.**

Two caveats on this result

Part of the GLITCH gain is available without splitting. If the artefact is purely seed- or initialisation-dependent, re-running the estimator with different seeds on the *same* data catches it at no cost in reach. What a seed jackknife cannot decorrelate is a defect tied to a specific *data* fluctuation — overfitting, or a fit driven by the actual events in that window. Only a data split touches those, which is exactly the data-directed failure mode, so the claim is narrower than the raw $+2.5\sigma$ but lands on the case at hand.

Second, and more important: after the split the residual spurious rate is essentially *entirely* coherent. The glitch and variance components contribute 0.9% and 0.1% of what still gets through. The split does not make the analysis safe — it changes what one has to worry about, from “is my estimator glitchy?” to “is my background model wrong in that window?”. That residual has to be controlled by other means (validation regions, spurious-signal tests, alternative background parameterisations), because the confirmation stage will endorse a mismodelled background as enthusiastically as a real signal.

The overall repricing is then a change of argument rather than of arithmetic. Section 6.3 stands exactly as written: with a perfect estimator, splitting relocates the look-elsewhere effect and cannot reduce it. But that was answering the wrong question for a data-directed search.

The split is not a look-elsewhere device — it is an **estimator-robustness device**, and priced as one, 0.5σ is cheap.

6.8 Recommended working point

If the scheme is adopted, the design that follows from the above is:

- exploration fraction $f \approx 0.25\text{--}0.35$, stage-1 threshold $Z_{\text{cut}} = 3$ (giving ≈ 9 pre-registered windows at this budget);
- pre-registered $\pm 2\sigma_M$ windows, with the outcome ladder written down before unblinding;
- the background model **trained on A alone** and applied frozen to B (arrangement 3 of Section 6.7), so that the confirmation significance comes from an estimator that has never seen a B event, without leaking B into the pre-registration;
- the claim quoted as a **global** significance over the pre-registered windows: evidence at $Z_B^{\text{global}} \geq 3$ (local $Z_B \geq 4.0$), discovery at $Z_B^{\text{global}} \geq 5$ (local 5.6). The false-evidence rate per opening is then 1.35×10^{-3} by construction, whatever number of windows A flags;
- B alone for the claim p -value, A+B for the measurement.

7 Software and reproducibility

Every number, table and figure in this note is produced by code in the accompanying repository, and the intention is that a reader can regenerate any of them rather than take them on trust. Nothing in the chain reads an experiment-internal input, so “regenerate” means from a clone and nothing else.

7.1 The two modules everything rests on

The whole budget derives from two data-free modules, and keeping them small and singular is what makes the claim checkable:

- `scripts/bump_observables.py` — the bump observables with their fractional resolutions, analyzable floors and published scan windows, each window carrying a per-channel source note (Appendix A). It also owns the two structural maps that decide what counts as one spectrum: the same-axis merges and the lepton-flavour split. It is imported, never duplicated; a floor or a window redefined in a consumer would be a bug.
- `scripts/public_obs_map.py` — the map from public BSM model classes onto the spectra they populate, plus the published event-selection multiplicity per spectrum.

Both carry import-time assertions that keep the flavour layer complete: every leaf channel must have a window and a floor, and no flavour-inclusive parent may keep one, which is what prevents a channel being counted both as itself and as part of its parent.

7.2 Layout and rebuilding

- `scripts/`, the pipeline, one concern per file.
- `results/tables/`, machine-readable output, including the cached Monte Carlo of Section 5.
- `results/plots/`, the figures of this note.
- `results/overviews/`, the long-form reports the sections here condense.

- `docs/`, holding `METHOD_NOTES.md` (conventions and the reasoning behind the counting choices), `OUTPUTS.md` (every output file mapped to the script that writes it), and this note.

The pipeline is driven by a single makefile whose targets are real files with real prerequisites, so a rerun redoes only what is stale:

```

pip install -r requirements.txt    # numpy, scipy, matplotlib
make all                          # budget + ab + stats
make help                         # the individual stages

```

The stages are `budget` (Sections 2 to 4), `ab` (Section 6) and `stats` (Section 5). The `budget` itself uses only the Python standard library; `ab` and `stats` need NumPy [27], SciPy [28] and Matplotlib [29].

7.3 Determinism

All Monte Carlo is explicitly seeded, and rebuilding the figure set reproduces the committed files byte for byte. That is worth more than it sounds, because it means a changed figure signals that an input or a method changed rather than that a random seed moved. The two genuinely slow computations, the Benjamini–Hochberg q scan and its imperfect-estimator variant, are cached as `.npz` files and rerun only on request.

7.4 Relation to the catalogue note

One question is deliberately absent here: which of these 46 spectra a given collaboration has actually produced signal samples for. That is a statement about a production program rather than about the search budget, it cannot be made from public information, and answering it does not change any number in this note — the windows are published-search windows either way. It is treated in a companion ATLAS-internal note, which imports the two modules above unchanged so that the two documents cannot drift apart on what a spectrum is.

8 Summary and recommendations

8.1 What was established

1. **Models collapse onto spectra.** Classified by *where the bump appears* rather than by theory, and with lepton flavour and b -tag content counted as part of the mass axis, the public BSM resonance model classes map onto **46 canonical spectra**. The number of searches is set by the number of spectra, not by the number of models.
2. **The search budget.** $N_{\text{trials}} \approx 3.7 \times 10^3$ inclusively, 6.6×10^3 at realistic event-selection granularity, hence $Z_{\text{local}} \approx 6.4$ to 6.5 for a 5σ global discovery, stable to $\pm 0.1\sigma$ against every counting choice varied. **Covering the entire known BSM resonance space costs about one extra unit of σ over a single counting experiment.**
3. **The combinatorial scan costs half a sigma more.** 150 exclusive object categories and 1094 mass groups give $N_{\text{trials}} = 1.4 \times 10^5$ and $Z_{\text{local}} = 6.98$.
4. **Most of the reachable space has never been scanned.** 57 of the 82 reachable object compositions have no published ATLAS bump hunt, under a deliberately generous mapping. Two-body space is saturated, while three- and four-body space is 75 to 90 % untouched. The ATLAS general search declined that coverage explicitly on look-elsewhere grounds, a trade the numbers above reprise.

5. **The arithmetic is self-consistent.** The same trials-factor calculation post-dicts the observed population of ATLAS excesses: tens of 3σ , all faded, and no spurious 5σ .
6. **Rank-based candidate selection is dominated.** At an identical false-alarm budget, a fixed threshold confirms more signal than the argmax at every signal strength, and a Benjamini–Hochberg rule lands between them. With an imperfect significance estimator the argument gets stronger, not weaker: the threshold’s margin grows to more than ten percentage points and the argmax loses control of its false-alarm rate entirely.
7. **A/B splitting relocates the look-elsewhere effect rather than reducing it.** The cost factorises exactly, and the residual two-coin penalty is $+0.3$ to $+0.5\sigma$ at any trials count. Symmetrising the scheme — running both directions and claiming on the union — does not recover it: it repairs a 50/50 split but is a wash against the asymmetric optimum, and it forfeits procedural blindness. The scheme is worth adopting when the trials factor one would have to defend exceeds roughly $13\times$ the true one, which is the situation a machine-learning-driven scan creates.

8.2 Recommendations

For an analysis. Quote the global significance against a counted trials factor and state the counting convention (published scan windows, resolution elements) explicitly. That is what makes 6.4 rather than 5 the number to beat, and it is defensible in a way that “we scanned a lot of spectra” is not. Select stage-1 candidates on absolute significance at a fixed expected-fake budget rather than by rank, and quote the *achieved* false-alarm rate rather than a nominal FDR parameter. If a two-stage scheme is adopted, use an asymmetric split ($f \approx 0.25$ to 0.35), pre-register the windows and the outcome ladder, use cross-half trainings, and expect to pay 0.5σ for the auditability.

For the search program. The three- and four-body mixed-flavour space is the largest identified gap, and it is not expensive to close: moving from the published program to a full combinatorial scan costs 0.5σ in the discovery bar. Signatures scanned once at Run-1 luminosity and then abandoned, such as $m(Zb)$, cost nothing in trials to revisit, because Section 2 has already counted their axes.

For a sample program. The exchange rate is unambiguous, whoever is doing the prioritising. Adding a model to an already-scanned spectrum costs *zero* trials, while opening a new spectrum costs its n_s — from 14 looks for $m(Ht)$ to 411 for $m(\gamma\gamma)$ (Table 1). Any argument about which channels to invest in should be conducted in those units.

8.3 Known limitations

The fractional resolutions are coarse per-channel central values, which is why every headline carries a factor-two band. Scan windows are hand-curated from published search families and inevitably involve judgement at the edges. The trials factor for the full ATLAS program used in Section 4 spans a factor ten and is the dominant uncertainty there, although the conclusion survives the whole range. Cross-channel correlations are neglected, which makes the sum mildly conservative, and event selections within one search family are partly correlated and often statistically combined, so the selections-level count is an over-count rather than a refinement. The model-to-spectrum map is assembled from public model databases and the published search record; it is complete to the best of our knowledge but is not machine-verifiable, and a missing model class that populates only an already-counted spectrum would not change N_{trials} at all.

A Scan windows and their sources

The windows below are the single input that fixes the search budget of Section 2. Each is the mass range a real, published ATLAS search family scans on that axis, extended where that family’s published signal grid demonstrably prepares a wider scan. They live in one module in the repository (`scripts/bump_observables.py`) so that every plot and table keyed on a bump observable uses the same edges; the source column reproduces the per-channel note recorded there.

Table 15: Published-search scan windows per bump observable.

observable	window [GeV]	basis
$m(\gamma\gamma)$	60–110, 150–5000	low-mass diphoton 66–110; spin-0/2 high mass to 5 TeV (incl. $X \rightarrow aa \rightarrow 4\gamma$)
$m(ee)$	150–8000	high-mass Drell–Yan dielectron scan (signal grid prepared to 8 TeV)
$m(\mu\mu)$	150–8000	high-mass Drell–Yan dimuon scan (signal grid prepared to 8 TeV)
$m(ee) (Z_d)$	1–400	prompt dark-photon e -jets 1–10; $H \rightarrow Z_d Z_d \rightarrow 4e$ 1–60; heavy $Z_d Z_d$ grid to 400
$m(\mu\mu) (Z_d)$	0.3–400	prompt dark-photon muon-jets 0.3–10; $H \rightarrow Z_d Z_d \rightarrow 4\mu$ 1–60; grid to 400
$m(e\mu)$ LFV	100–8000	$Z' \rightarrow e\mu$ (grid 100 GeV–8 TeV)
$m(e\tau), m(\mu\tau)$ LFV	100–3000	$Z' \rightarrow e\tau/\mu\tau$ (hadronic-tau channel, grid to 3 TeV)
$m(\ell^\pm \ell'^\pm)$ SS ($ee, e\mu, \mu\mu$)	200–1300	$H^{\pm\pm}$ same-sign dilepton pair scan, per flavour pair
multilepton	50–10000	type-III seesaw / VLL / prompt heavy- N mass hypotheses
$m(ej), m(\mu j)$	200–5000	leptoquark pair $\rightarrow \ell q$ [6]; QBH jet-electron/jet-muon
$m(\tau j)$	200–5000	leptoquark pair $\rightarrow \tau q$ (LQ3)
$m(cb)$ dijet	20–1000	$t \rightarrow H^+(cb)b$ 60–160; tb -associated heavy $H^+ \rightarrow cb$ to 1 TeV
$m(jj)$	200–8000	dijet: TLA+ISR 200–450, TLA 450–1800, high mass to 8 TeV
$m(bj)$	200–6000	$b^* \rightarrow bg$, b -tagged dijet resonance [5]
$m(3j)$	200–1800	RPV gluino $\rightarrow$ three-jet paired resonance scan
$m(bb)$	15–62, 450–6000	$h \rightarrow aa \rightarrow 4b$ low mass 15–62; b -tagged dijet 450 GeV–6 TeV
$m(j\gamma)$	500–7000	$q^* \rightarrow q\gamma$ (grid 500 GeV–7 TeV)
$m(t\bar{t})$	350–6000	$t\bar{t}$ resonance, resolved and boosted, from the $2m_t$ threshold
$m(t\bar{t})/m(jj)$	400–2000	pair-produced coloron/sgluon, per-particle $t\bar{t}$ and jj legs
$m(tb)$	180–6000	$W' \rightarrow tb$ and heavy $H^+ \rightarrow tb$
$m(VV)$	200–6000	$WW/WZ/ZZ$ diboson family, ggF and VBF
$m(Vh)$	200–5000	$W/Z + h(bb)$ resonance

Table 15 continued

observable	window [GeV]	basis
$m(HH)$	250–6000	$X \rightarrow HH$ in $bbbb$, $bb\tau\tau$, $bb\gamma\gamma$, including boosted $4b$
$m(t\bar{t}Z)/m(Zt)$	1000–4000	VLQ $T \rightarrow Zt$ single and pair production; $W' \rightarrow V_{lt}$ cascade
$m(\text{multi})$	3000–10500	quantum black hole / sphaleron multi-object threshold scan
$m(eejj)$, $m(\mu\mu jj)$	400–7000	$W_R \rightarrow \ell\ell jj$ (Keung–Senjanović)
$m_T(e\nu)$, $m_T(\mu\nu)$	150–7000	$W' \rightarrow e\nu/\mu\nu$ transverse mass
$m_T(\tau\nu)$	200–5000	$W' \rightarrow \tau\nu$
$m(\tau\tau)$	35–6000	$a/H/A/Z' \rightarrow \tau\tau$ (low-mass grid from 35 GeV)
<i>channels whose scan is published, or was published below 13 TeV</i>		
$m(V\gamma)$	220–6000	$X \rightarrow Z\gamma/W\gamma$ ($\ell\ell$ or qq plus photon)
$m(e\gamma)$, $m(\mu\gamma)$	200–5000	excited lepton $\ell^* \rightarrow \ell\gamma$; 7 and 8 TeV only [9, 10]
$m(eZ)$, $m(\mu Z)$	100–1100	RPV chargino/neutralino trilepton $\ell + Z$ resonance scan [30]
$m(eb)$, $m(\mu b)$	200–2000	RPV $\tilde{t} \rightarrow b\ell$ [31]; b/c -tagged leptoquark pair leg
$m(\tau b)$	200–2000	third-generation leptoquark $\rightarrow \tau b$
$m(tW)$	500–3000	excited $b^* \rightarrow tW$
$m(Wb)$	800–3000	single VLQ $T/Y \rightarrow Wb$
$m(Ht)$	1000–3000	single VLQ $T \rightarrow Ht$

One documented near-miss and one documented omission. The $m(\ell Z)$ scan was long booked as “8 TeV only [17], absorbed into the multilepton axis” – wrongly: the Run-2 trilepton resonance scan [30] covers $e + Z$ and $\mu + Z$ as separate spectra, and they now stand in the budget as $m(eZ)/m(\mu Z)$. The $m(Zb)$ scan, performed once at 8 TeV [18] with no Run-2 successor, remains omitted; including it would shift N_{trials} by less than one percent.

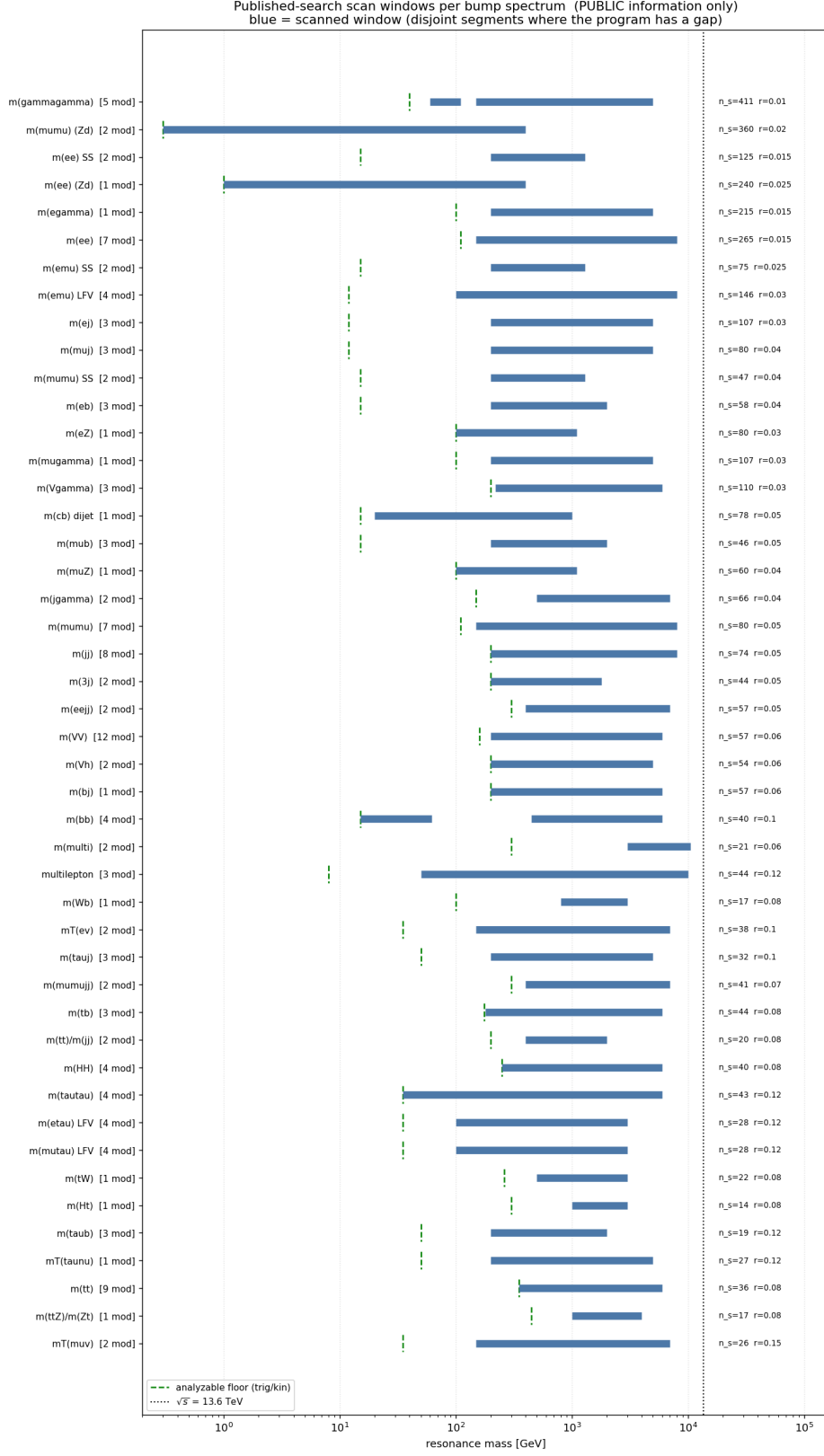


Figure 14: The scan windows of Table 15, drawn per spectrum, with the analyzable floor of each observable marked. Spectra scanned in disjoint segments ($m(\gamma\gamma)$, $m(bb)$) show both.

References

- [1] Sergey Volkovich, Federico De Vito Halevy, and Shikma Bressler. A Data-Directed Paradigm for BSM searches: the bump-hunting example. *Eur. Phys. J. C*, 82:265, 2022. doi: 10.1140/epjc/s10052-022-10215-1. arXiv:2107.11573.
- [2] ATLAS Collaboration. Search for new phenomena in two-body invariant mass distributions using unsupervised machine learning for anomaly detection at $\sqrt{s} = 13$ TeV with the ATLAS detector. *Phys. Rev. Lett.*, 132:081801, 2024. doi: 10.1103/PhysRevLett.132.081801. arXiv:2307.01612.
- [3] ATLAS Collaboration. A strategy for a general search for new phenomena using data-derived signal regions and its application within the ATLAS experiment. *Eur. Phys. J. C*, 79:120, 2019. doi: 10.1140/epjc/s10052-019-6540-y. arXiv:1807.07447.
- [4] Eilam Gross and Ofer Vitells. Trial factors for the look elsewhere effect in high energy physics. *Eur. Phys. J. C*, 70:525, 2010. doi: 10.1140/epjc/s10052-010-1470-8. arXiv:1005.1891.
- [5] ATLAS Collaboration. Search for heavy particles in the b -tagged dijet mass distribution with additional b -tagged jets in proton-proton collisions at $\sqrt{s} = 13$ TeV with the ATLAS experiment. *Phys. Rev. D*, 105:012001, 2022. doi: 10.1103/PhysRevD.105.012001. arXiv:2108.09059.
- [6] ATLAS Collaboration. Search for dijet resonances in events with an isolated charged lepton using $\sqrt{s} = 13$ TeV proton-proton collision data collected by the ATLAS detector. *JHEP*, 06:151, 2020. doi: 10.1007/JHEP06(2020)151. arXiv:2002.11325.
- [7] ATLAS Collaboration. Search for new phenomena in multi-body invariant masses in events with at least one isolated lepton and two jets using $\sqrt{s} = 13$ TeV proton-proton collision data collected by the ATLAS detector. *JHEP*, 07:202, 2023. doi: 10.1007/JHEP07(2023)202. arXiv:2211.08945.
- [8] ATLAS Collaboration. A search for new resonances in multiple final states with a high transverse momentum Z boson in $\sqrt{s} = 13$ TeV pp collisions with the ATLAS detector. *JHEP*, 06:036, 2023. doi: 10.1007/JHEP06(2023)036. arXiv:2209.15345.
- [9] ATLAS Collaboration. Search for excited leptons in proton-proton collisions at $\sqrt{s} = 7$ TeV with the ATLAS detector. *Phys. Rev. D*, 85:072003, 2012. doi: 10.1103/PhysRevD.85.072003. arXiv:1201.3293.
- [10] ATLAS Collaboration. Search for excited electrons and muons in $\sqrt{s} = 8$ TeV proton-proton collisions with the ATLAS detector. *New J. Phys.*, 15:093011, 2013. doi: 10.1088/1367-2630/15/9/093011. arXiv:1308.1364.
- [11] ATLAS Collaboration. Search for doubly-charged Higgs bosons in like-sign dilepton final states at $\sqrt{s} = 7$ TeV with the ATLAS detector. *Eur. Phys. J. C*, 72:2244, 2012. doi: 10.1140/epjc/s10052-012-2244-2. arXiv:1210.5070.
- [12] ATLAS Collaboration. Search for doubly charged Higgs boson production in multi-lepton final states using 139 fb^{-1} of proton-proton collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector. *Eur. Phys. J. C*, 83:605, 2023. doi: 10.1140/epjc/s10052-023-11578-9. arXiv:2211.07505.
- [13] ATLAS Collaboration. Search for resonant pair production of Higgs bosons in the $b\bar{b}b\bar{b}$ final state using pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector. *Phys. Rev. D*, 105:092002, 2022. doi: 10.1103/PhysRevD.105.092002. arXiv:2202.07288.

- [14] CMS Collaboration. Search for resonances in events with photon and jet final states in proton-proton collisions at $\sqrt{s} = 13$ TeV. *JHEP*, 12:189, 2023. doi: 10.1007/JHEP12(2023)189. arXiv:2305.07998.
- [15] ATLAS Collaboration. Search for excited τ -leptons and leptoquarks in the final state with τ -leptons and jets in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector. *JHEP*, 06:199, 2023. doi: 10.1007/JHEP06(2023)199. arXiv:2303.09444.
- [16] ATLAS Collaboration. Search for same-charge top-quark pair production in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector. *JHEP*, 02:084, 2025. doi: 10.1007/JHEP02(2025)084. arXiv:2409.14982.
- [17] ATLAS Collaboration. Search for heavy lepton resonances decaying to a Z boson and a lepton in pp collisions at $\sqrt{s} = 8$ TeV with the ATLAS detector. *JHEP*, 09:108, 2015. doi: 10.1007/JHEP09(2015)108. arXiv:1506.01291.
- [18] ATLAS Collaboration. Search for pair and single production of new heavy quarks that decay to a Z boson and a third-generation quark in pp collisions at $\sqrt{s} = 8$ TeV with the ATLAS detector. *JHEP*, 11:104, 2014. doi: 10.1007/JHEP11(2014)104. arXiv:1409.5500.
- [19] S. V. Chekanov. On the Statistical Interpretation of Discoveries in LHC Data. 2026. preprint, arXiv:2605.24441.
- [20] Pouya Asadi et al. Digging Deeper for New Physics in the LHC Data. *JHEP*, 11:194, 2017. doi: 10.1007/JHEP11(2017)194. arXiv:1707.05783.
- [21] Louis Lyons. Open statistical issues in particle physics. *Ann. Appl. Stat.*, 2:887, 2008. doi: 10.1214/08-AOAS163. arXiv:0811.1663.
- [22] ATLAS Collaboration. Summary plots and public results of ATLAS searches. <https://twiki.cern.ch/twiki/bin/view/AtlasPublic/SearchesPublicResults>.
- [23] R. A. Fisher and L. H. C. Tippett. Limiting forms of the frequency distribution of the largest or smallest member of a sample. *Math. Proc. Camb. Phil. Soc.*, 24:180, 1928. doi: 10.1017/S0305004100015681.
- [24] E. J. Gumbel. *Statistics of Extremes*. Columbia University Press, 1958.
- [25] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J. R. Stat. Soc. B*, 57:289, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.
- [26] H. E. Daniels. The statistical theory of the strength of bundles of threads. I. *Proc. R. Soc. Lond. A*, 183:405–435, 1945. doi: 10.1098/rspa.1945.0011. source of the identity $P(\sup_t F_n(t)/t \geq 1/q) = q$ for the uniform empirical distribution, from which the exact null behaviour of the step-up procedure follows.
- [27] Charles R. Harris et al. Array programming with NumPy. *Nature*, 585:357, 2020. doi: 10.1038/s41586-020-2649-2.
- [28] Pauli Virtanen et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261, 2020. doi: 10.1038/s41592-019-0686-2.
- [29] John D. Hunter. Matplotlib: A 2D graphics environment. *Comput. Sci. Eng.*, 9:90, 2007. doi: 10.1109/MCSE.2007.55.

- [30] ATLAS Collaboration. Search for trilepton resonances from chargino and neutralino pair production in $\sqrt{s} = 13$ TeV pp collisions with the ATLAS detector. *Phys. Rev. D*, 103: 112003, 2021. doi: 10.1103/PhysRevD.103.112003. arXiv:2011.10543.
- [31] ATLAS Collaboration. A search for R -parity violating supersymmetric decays of the top squark to a b -jet and a lepton in $\sqrt{s} = 13$ TeV pp collisions with the ATLAS detector. *Phys. Rev. D*, 110:092004, 2024. doi: 10.1103/PhysRevD.110.092004. arXiv:2406.18367.